Chapter 3

Questions on Lesson Four

- The Electric Transformer.
- The Electric Motor.

Interactive test



The questions signed by ✱ are answered in detail. Understand Apply Higher Order Thinking Skills

Important guidelines

The electric transformer

Ideal

- $\frac{V_s}{V_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$
- $(P_w)_p = (P_w)_s \quad , \quad V_p I_p = V_s I_s$
- In the case of two secondary coils:
  - When each device operates separately:

    $\frac{V_p}{(V_s)_1} = \frac{N_p}{(N_s)_1} \quad , \quad \frac{V_p}{(V_s)_2} = \frac{N_p}{(N_s)_2}$
  - When the two devices work together at the same time:

    $(P_w)_p = (P_w)_{s_1} + (P_w)_{s_2}$

Non-ideal

- $\eta = \frac{(P_w)_s}{(P_w)_p} \times 100 = \frac{V_s I_s}{V_p I_p} \times 100$

  $= \frac{V_s N_p}{V_p N_s} \times 100$
- $(P_w)_p > (P_w)_s$
- In the case of two secondary coils:

  $\eta\,(P_w)_p = \left((P_w)_{s_1} + (P_w)_{s_2}\right) \times 100$

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- The lost power in the wires $= I^2_{eff} R$
- Drop in voltage $= I_{eff} R$
- Power at the consumer = Power at station – Lost power in the wires
- The efficiency of transfer $= \frac{\text{The power at the distribution regions}}{\text{The power at the station}} \times 100$

First Multiple choice questions

Electric transformer

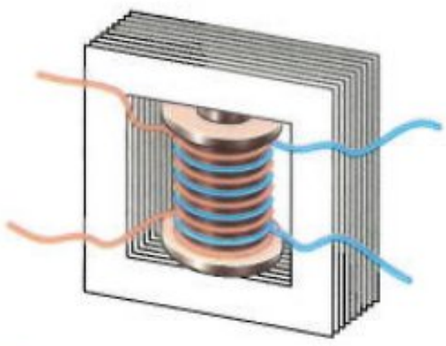
1. The electric transformer doesn't work when the current passing through its primary coil is ..... .

(a) unidirectional of variable intensity (b) alternating current
(c) unidirectional of constant intensity (d) having high frequency

2. Which of the following choices describes the parts of the step-up transformer?

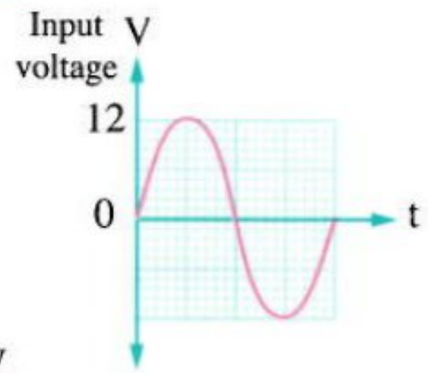
	Input voltage	Core	Primary coil	Secondary coil
(a)	DC	Solid	100 turns	10 turns
(b)	DC	Soft iron	10 turns	100 turns
(c)	AC	Soft iron	100 turns	10 turns
(d)	AC	Soft iron	10 turns	100 turns

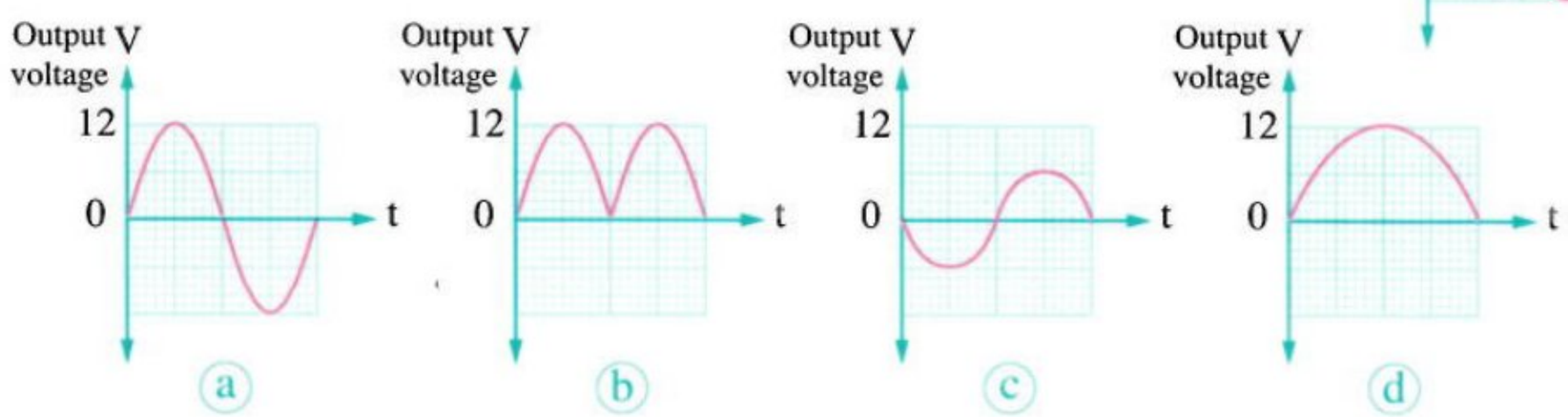
3. The electric transformer is designed such that the turns of its coils, the primary and the secondary are insulated and interwined as shown in the opposite figure and that reduces the loss in the electric energy caused by ..... .



(a) eddy currents (b) electric resistance
(c) magnetic flux leakage (d) self-induction

4. The opposite figure shows the input voltage of a step-down transformer for a certain period of time, therefore the figure of the output voltage in the same period of time is ..... .





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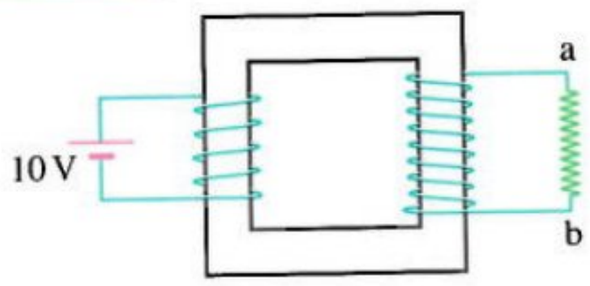
5 The physical quantity that decreases in the secondary coil of an ideal step-up transformer than its value in the primary coil is ..... .

(a) the electric power
(b) the maximum value of the electric current
(c) the current frequency
(d) the magnetic flux

6 The physical quantity that increases in the secondary coil of an ideal step-down transformer is ..... .

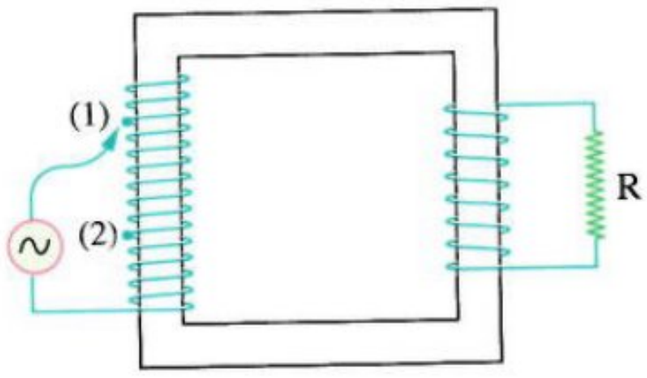
(a) the electric power
(b) the current intensity
(c) the current frequency
(d) the magnetic flux

7 In the opposite figure, the potential difference between points a, b is ..... .



(a) less than 10 V
(b) greater than 10 V
(c) equal to 10 V
(d) equal to 0

8 The opposite figure represents an ideal electric transformer, if the slider is moved from point (1) to point (2), then :



	The value of current in the secondary coil	The electric power drawn from an AC source
(a)	Increases	Increases
(b)	Increases	Decreases
(c)	Decreases	Increases
(d)	Decreases	Decreases

9 * An ideal transformer draws an input power of 300 watts when the current of its secondary coil is 5 A, then the potential difference across its secondary coil equals ..... .

(a) 20 V
(b) 40 V
(c) 60 V
(d) 100 V

10 It's required to use an ideal transformer to raise the voltage from 10 V to 50 V, if the number of turns of the primary coil is 80 turns, then the number of turns of the secondary coil equals ..... .

(a) 400 turns
(b) 250 turns
(c) 200 turns
(d) 150 turns

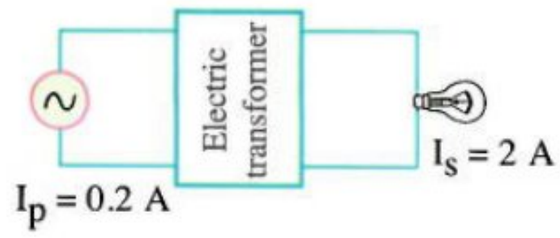
11 A step-down electric transformer of efficiency 90% has two coils where the ratio between their number of turns is 33 : 2, so if the voltage across the terminals of its secondary coil is 12 V, the source voltage equals ..... .

(a) 440 V
(b) 244 V
(c) 220 V
(d) 110 V

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12 ✱ The opposite figure shows an ideal electric transformer connected between an AC source and a bulb, then the type of the transformer and the ratio between number of turns of its two coils $(\frac{N_s}{N_p})$ are ..... .



(a) step-down, $\frac{10}{1}$ (b) step-down, $\frac{1}{10}$ (c) step-up, $\frac{1}{10}$ (d) step-up, $\frac{10}{1}$

13 ✱ An ideal transformer works on a primary potential difference of 240 V, if the number of turns of the secondary coil is double that of the primary coil and the value of the current of the primary coil is 3 A, then:

(i) The potential difference across the terminals of the secondary coil equals ..... .

(a) 480 V (b) 240 V (c) 160 V (d) 120 V

(ii) The value of the electric current in the secondary coil equals ..... .

(a) 6 A (b) 4.5 A (c) 3 A (d) 1.5 A

(iii) The produced electric power equals ..... .

(a) 720 W (b) 360 W (c) 320 W (d) 240 W

14 ✱ In an ideal step-down transformer, the number of turns of the primary coil is 5000 turns and the number of turns of the secondary coil is 250 turns.

If the voltage of its primary coil is 240 V, then:

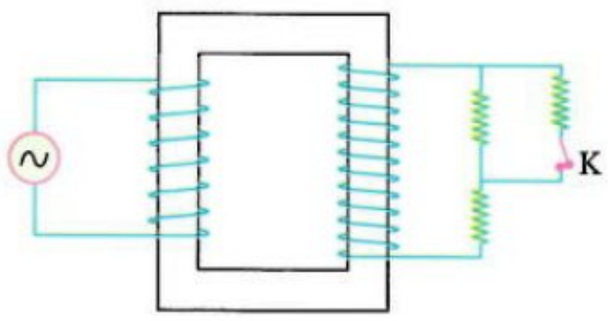
(i) The induced emf across the terminals of its secondary coil equals ..... .

(a) 12 V (b) 7 V (c) 6 V (d) 3 V

(ii) The ratio between the current of the primary coil to that of the secondary coil $(\frac{I_p}{I_s})$ equals ..... .

(a) $\frac{1}{20}$ (b) $\frac{20}{1}$ (c) $\frac{25}{24}$ (d) $\frac{24}{25}$

15 The opposite figure represents an ideal electric transformer, after opening the switch (K), then:



(i) The potential difference across the terminals of the secondary coil ..... .

(a) increases (b) decreases

(c) remains unchanged (d) the answer is indeterminable

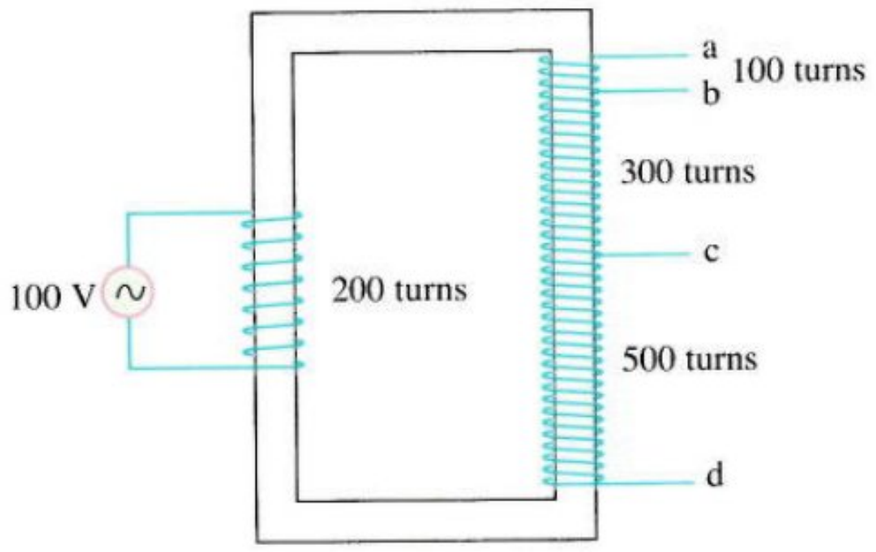
(ii) The value of the current passing through the primary coil ..... .

(a) increases (b) decreases

(c) remains unchanged (d) the answer is indeterminable

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16 The opposite figure represents an ideal electric transformer to be used in operating an electric appliance, what are the two points at which the appliance can be connected if it works at:



(i) a potential difference of 50 V?

(a) a, b (b) a, c
(c) b, d (d) a, d

(ii) a potential difference of 400 V?

(a) a, c (b) b, c (c) b, d (d) c, d

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17 An ideal electric transformer is connected to an AC source of 50 V and raises the voltage by an amount of 25 V. When 40 turns are removed from the secondary coil and the remaining turns are connected to the transformer, it starts reducing the voltage by an amount of 25 V, so the number of turns of the primary coil is ..... .

(a) 20 (b) 40 (c) 60 (d) 80

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18 A step-up transformer is used to raise the potential difference from 120 V to 3000 V where the electric current passing in its primary coil is 2 A and the current passing in its secondary coil is 0.06 A, the efficiency of that transformer equals ..... .

(a) 75% (b) 80% (c) 85% (d) 100%

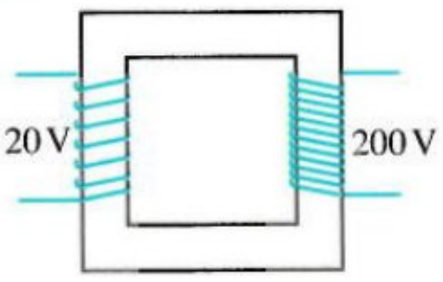
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19 * An electric transformer converts the AC voltage from 220 V into 17.6 V where the ratio between the number of turns of its coils is 10 : 1, so the efficiency of the transformer equals ..... .

(a) 80% (b) 40% (c) 8% (d) 0.8%

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20 The opposite figure shows a step-down transformer of efficiency 80%, if the number of turns of the primary coil is 640 turns, then the number of turns of the secondary coil equals ..... .



(a) 80 turns (b) 70 turns (c) 40 turns (d) 20 turns

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21 The efficiency of an electric transformer is 80% and that means ..... .

(a) the power loss through it is 80%
(b) the power of the secondary coil is 20% of that of the primary coil
(c) the power loss through it is 20%
(d) the power of the primary coil is 80% of that of the secondary coil

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22 An electric transformer has an efficiency of 80%, its primary coil is connected to an AC source of power 40 kW, then the power of the secondary coil equals ..... .

(a) 38 kW (b) 8 kW (c) 48 kW (d) 32 kW

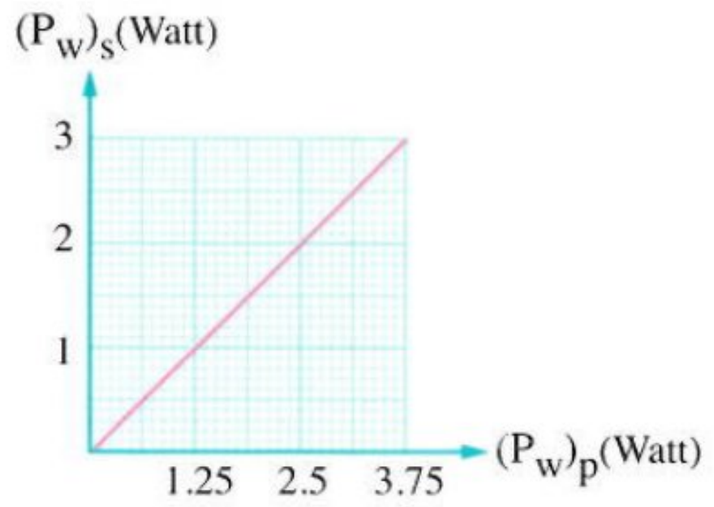
23 The ratio between the number of turns of the secondary coil to the number of turns of the primary coil of an ideal electric transformer is equal to $\frac{3}{2}$. If the output power of the secondary coil of the transformer = P_w, then the input power of the primary coil equals ..... .

(a) P_w (b) $1.5\ P_w$ (c) $\frac{2}{3}\ P_w$ (d) $5\ P_w$

24 An electric transformer of efficiency 96% has two coils, the ratio between their number of turns is $\frac{N_p}{N_s} = \frac{8}{5}$, then the ratio between the intensities of the electric currents that flow through the two coils of the transformer $\left(\frac{I_p}{I_s}\right)$ is ..... .

(a) $\frac{8}{5}$ (b) $\frac{3}{5}$ (c) $\frac{5}{3}$ (d) $\frac{5}{8}$

25 The opposite graph represents the relation between the power of the secondary coil $(P_w)_s$ and the power of the primary coil $(P_w)_p$ for a step-down transformer in which the ratio between the number of its coils turns is $\frac{1}{20}$, so:



(i) The efficiency of the transformer equals ..... .

(a) 100% (b) 95%

(c) 80% (d) 75%

(ii) The value of the current intensity in the primary coil if the current value in the secondary coil is 2 A equals ..... .

(a) 0.1 A (b) 0.2 A (c) 0.3 A (d) 0.4 A

26 A step-up transformer of efficiency 80%, the ratio between the number of turns of the primary coil to that of secondary coil is 1 : 16, so the ratio between the frequency of current in its primary coil and that in its secondary coil is ..... .

(a) 1 : 16 (b) 8 : 10 (c) 16 : 1 (d) 1 : 1

27 An electric transformer of efficiency 90% is connected to AC source of potential difference 110 volts. If the electric current intensities passing through its primary and secondary coils are 2 A and 18 A respectively, then:

(i) The output power of the secondary coil equals ..... .

(a) 120 W (b) 150 W (c) 167 W (d) 198 W

(ii) The ratio between the number of turns of the primary coil to the number of turns of the secondary coil equals ..... .

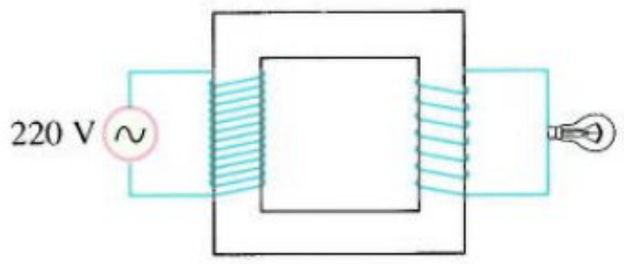
(a) $\frac{9}{1}$ (b) $\frac{1}{9}$ (c) $\frac{10}{1}$ (d) $\frac{1}{10}$

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28 An electric transformer of efficiency 96% is connected to 10 electric ovens that are in parallel with each other where each oven of them works on a potential difference of 220 V and draws an electric current of intensity 15 A, then the consumed electric power in the primary coil of the transformer equals ..... .

(a) 3.9×10^4 W (b) 3.8×10^4 W (c) 3.6×10^4 W (d) 3.4×10^4 W

29 In the opposite figure, an electric transformer of efficiency 96% has a primary coil of 440 turns and a secondary coil that is connected to an electric lamp of power 36 W that works on a potential difference of 24 V, so the number of turns of the secondary coil that matches the same specifications of the lamp is ..... .

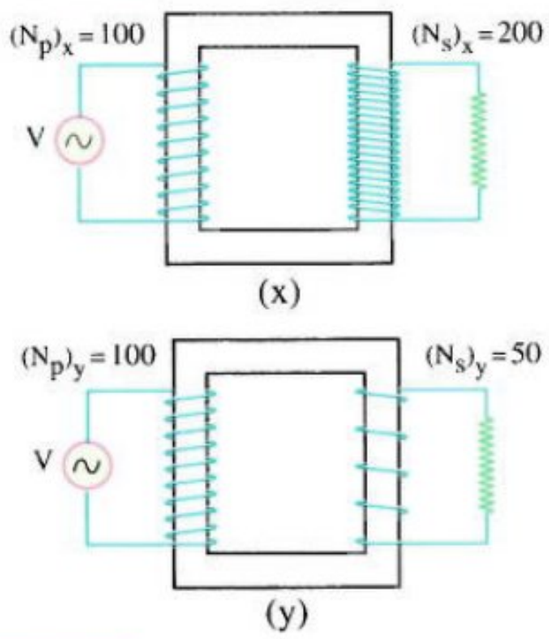


(a) 36 turns (b) 50 turns (c) 110 turns (d) 220 turns

30 An ideal electric transformer is connected to an AC source of voltage 30 V, if the potential difference across the terminals of each turn of the turns of the secondary coil is 0.3 V, then the number of turns of the primary coil equals ..... .

(a) 50 (b) 100 (c) 150 (d) 200

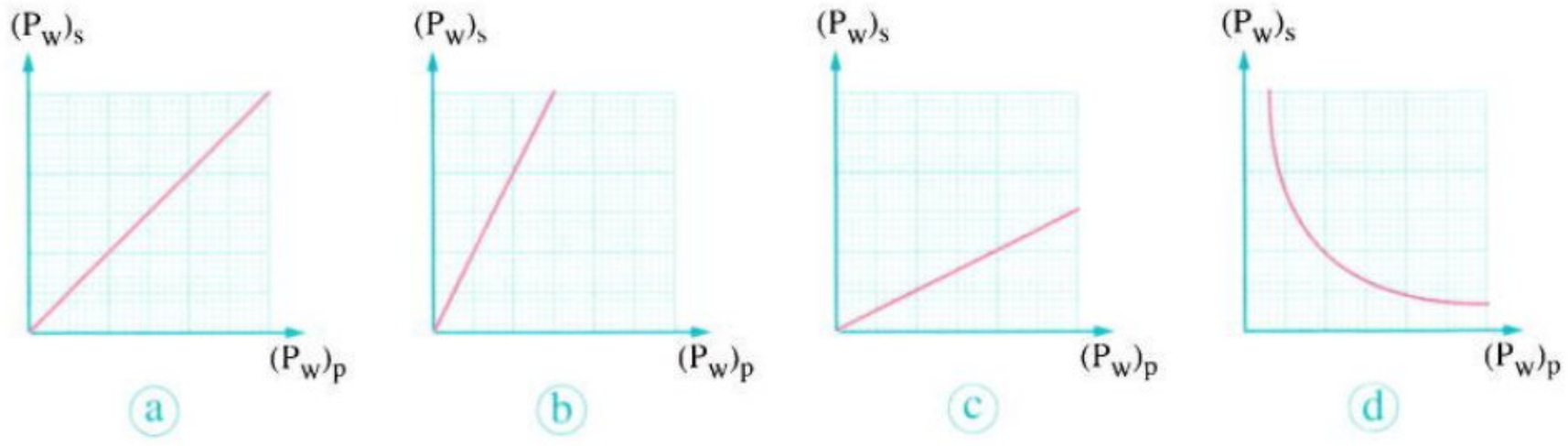
31 The opposite figure represents two electric transformers (x) and (y) whose efficiencies are 80% and 90% respectively. If each transformer is connected to a source of potential difference V, then the ratio of the potential difference across each one turn of the secondary coil of transformer (x) to that of transformer (y) is ..... .



(a) $\frac{1}{4}$ (b) $\frac{4}{1}$

(c) $\frac{8}{9}$ (d) $\frac{9}{8}$

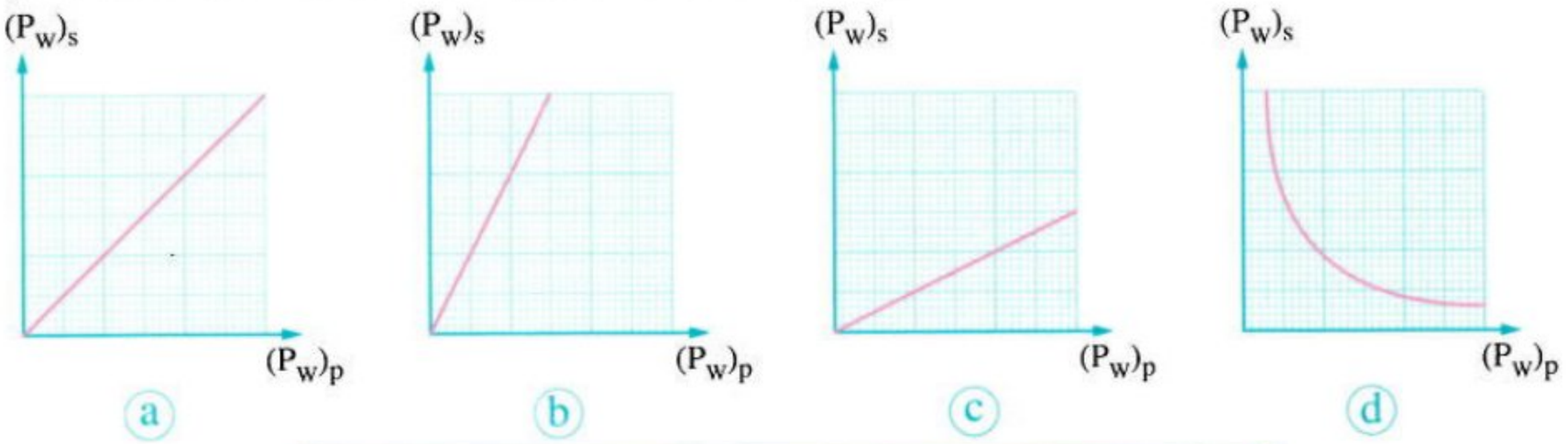
32 Which of the following graphs represents the relation between the power of the secondary coil $(P_w)_s$ and that of the primary coil $(P_w)_P$ in an ideal transformer? (Where the two axis are drawn in the same scale)



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33 Which of the following graphs represents the relation between the power of the secondary coil $(P_w)_s$ and that of the primary coil $(P_w)_P$ in a non-ideal transformer?
(Where the two axis are drawn in the same scale)



34 The number of turns of the two coils of an ideal step-down transformer are 100 and 80 turns, so if its secondary coil is connected to an electric device of power 1.5 kW that works at a potential difference of a maximum value of $60\sqrt{2}$ V, then the effective value of the current passing in the primary coil equals ..... .

(a) 31.25 A (b) 27.25 A (c) 22 A (d) 20 A

35 An electric transformer of efficiency 80% is connected to an AC power source of voltage effective value 240 volts. Given that the consumed power in the primary coil is 120 W and the number of turns of the primary coil is 3000 turns while the number of turns of the secondary coil is 1500 turns, then the type of the transformer and the current intensity passing through the secondary coil respectively are ..... , ...

(a) step-down transformer, 0.6 A (b) step-down transformer, 1 A
(c) step-up transformer, 1 A (d) step-up transformer, 0.6 A

36 ✱ A step-down transformer of efficiency 100% that has a secondary coil of 600 turns is connected to an electric source of emf 200 volts to operate a device of power 48 W at a potential difference of 24 volts, then:

(i) The number of turns of the primary coil equals ..... .

(a) 10000 turns (b) 5000 turns (c) 2500 turns (d) 1250 turns

(ii) The value of the current passing in the secondary coil equals ..... .

(a) 2 A (b) 1.5 A (c) 1 A (d) 0.5 A

(iii) The value of the current passing in the primary coil equals ..... .

(a) 0.06 A (b) 0.12 A (c) 0.18 A (d) 0.24 A

37 ✱ The ratio between the numbers of turns for the two coils in an ideal step-up transformer is 1 : 100. If the primary coil is connected to AC source of 200 volts, then:

(i) The induced emf in the secondary coil equals ..... .

(a) 2×10^4 V (b) 2×10^3 V (c) 2×10^2 V (d) 2 V

(ii) The ratio between the current value in the primary coil to that in the secondary coil respectively is ..... .

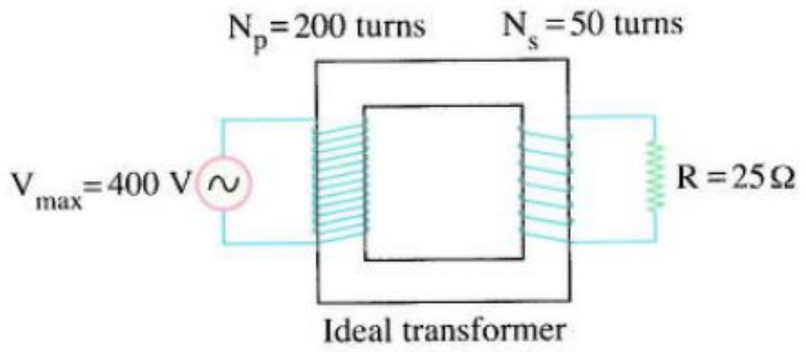
(a) $\frac{1}{200}$ (b) $\frac{1}{100}$ (c) $\frac{100}{1}$ (d) $\frac{200}{1}$

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(iii) The power produced in the secondary coil if the resistance of its circuit is 10 kΩ equals ..... .

(a) 4 W (b) 400 W (c) 4×10^3 W (d) 4×10^4 W

38 In the opposite figure, an ideal electric transformer has a primary coil connected to an AC source of maximum voltage 400 V, so the consumed electric power in the resistance R is approximately ..... .



(a) 100 W (b) 200 W (c) 300 W (d) 400 W

39 An electric transformer loses 36 kJ of its electric energy during an hour of its operation. If the potential difference across its secondary coil is 5 V and the value of the current passing through its secondary coil is 8 A, then the efficiency of the transformer equals ..... .

(a) 25% (b) 40% (c) 80% (d) 90%

40 * The secondary coil of an ideal electric transformer is connected to an electric bulb of resistance 10 Ω that consumes electric energy of 3000 J within 5 minutes, if the emf of the electric source that connected to the primary coil is 200 V, then:

(i) The effective value of the electric current passing in the primary coil equals ..... .

(a) 2.22×10^{-4} A (b) 0.013 A (c) 0.05 A (d) 3 A

(ii) The effective value of the electric current passing in the secondary coil equals ..... .

(a) 0.25 A (b) 1 A (c) 1.25 A (d) 2 A

(iii) The potential difference between the terminals of the secondary coil equals ..... .

(a) 2.5 V (b) 5 V (c) 10 V (d) 20 V

41 * Electric transformer is connected to AC source of 220 V, so a current of effective value 10 A passes through its primary coil. If the produced power in the secondary coil is 1980 W and the induced potential difference across its terminals is 22 V, then:

(i) The efficiency of the transformer equals ..... .

(a) 30% (b) 45% (c) 75% (d) 90%

(ii) The resistance of the secondary coil circuit equals ..... .

(a) 0.24 Ω (b) 0.12 Ω (c) 0.11 Ω (d) 0.01 Ω

42 * A step-down electric transformer of efficiency 90% has a primary coil voltage of 200 V and a secondary 9 V. If the intensity of the current in the primary is 0.5 A and the number of turns of the secondary is 90 turns, then the intensity of the current in the secondary coil and the number of turns of the primary coil respectively are ..... .

(a) 10 A, 1800 turns (b) 10 A, 900 turns (c) 5 A, 1800 turns (d) 5 A, 900 turns

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43 ✱ A step-down electric transformer of efficiency 80% is connected to an AC source of 2500 V which gives a current of 80 A in its secondary coil. The ratio between number of turns of the primary and the secondary coils is 20 : 1, so the emf across the two terminals of the secondary coil and also the intensity of the current in the primary coil respectively are ..... , ...

(a) 100 V, 4 A (b) 50 V, 2 A (c) 100 V, 2 A (d) 50 V, 4 A

44 ✱ An electric transformer of efficiency 80%, works on AC source of emf 200 V to give an emf of 8 V. If the number of turns of the primary coil is 1600 turns and the value of the current passing through it is 0.2 A, then:

(i) The number of turns of the secondary coil equals ..... .

(a) 100 turns (b) 80 turns (c) 60 turns (d) 20 turns

(ii) The lost electric power in the transformer equals ..... .

(a) 8 W (b) 25.6 W (c) 32 W (d) 40 W

(iii) The value of the current in the secondary coil equals ..... .

(a) 16 A (b) 5.3 A (c) 4 A (d) 3.2 A

45 ✱ An ideal electric transformer that has a primary coil of 1100 turns works on potential difference of 220 V. It has two secondary coils (x, y), coil x is connected to a small electric fan working on (6 V, 0.4 A) while coil y is connected to an electric bell working on (12 V, 0.35 A), then:

(i) The numbers of turns of both secondary coils (x, y) respectively are ..... .

(a) 30 turns, 60 turns (b) 20 turns, 60 turns

(c) 15 turns, 30 turns (d) 15 turns, 20 turns

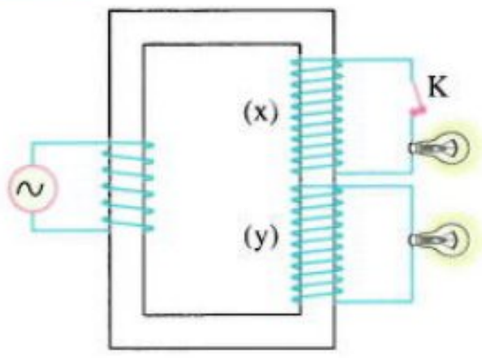
(ii) The electric current intensity of the primary coil when the fan and the bell are operated together equals ..... .

(a) 0.06 A (b) 0.04 A (c) 0.03 A (d) 0.015 A

46 ✱ TV set works on an AC potential difference whose maximum value is 550 V and its frequency is 50 Hz which is supplied by a step-up transformer whose primary coil is connected to an AC dynamo that has coil dimensions 20 cm, 10 cm rotating in a flux density of 0.14 tesla such that its number of turns equals half the number of turns of the primary coil of the transformer. Then the number of turns of the secondary coil of the transformer equals ..... . (Assuming that the efficiency of the transformer = 100%)

(a) 1250 turns (b) 625 turns (c) 312 turns (d) 12.5 turns

47 An ideal electric transformer has a primary coil of N turns and two secondary coils (x , y), each has 2 N turns and is connected to an electric lamp of resistance (R) as shown in the opposite figure, so after opening the switch (K), then:



(i) The brightness of the lamp connected to coil (y) ..... .

(a) increases (b) decreases but doesn't vanish

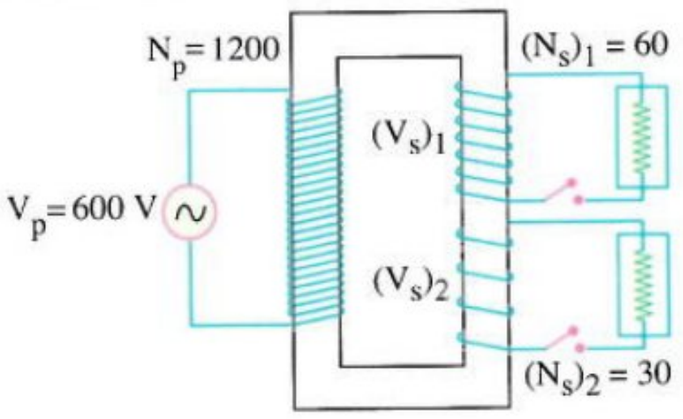
(c) vanishes (d) remains unchanged

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(ii) The electric power drawn from the AC source ..... .

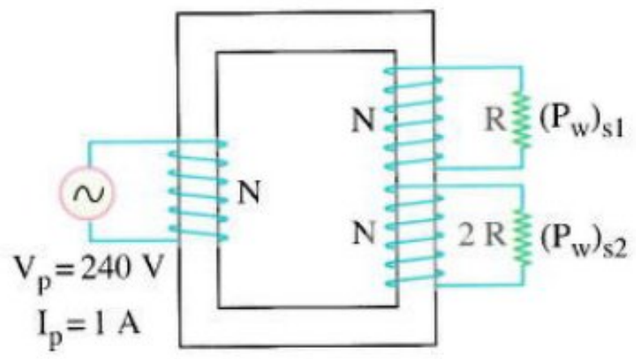
(a) increases to double
(b) decreases to its half
(c) increases to its four times
(d) decreases to its quarter

48 The opposite diagram represents an ideal transformer having two secondary coils that provide two electrical devices with electricity, so at operating each of them individually, their output voltages $(V_s)_1$ and $(V_s)_2$ respectively are ..... , ...



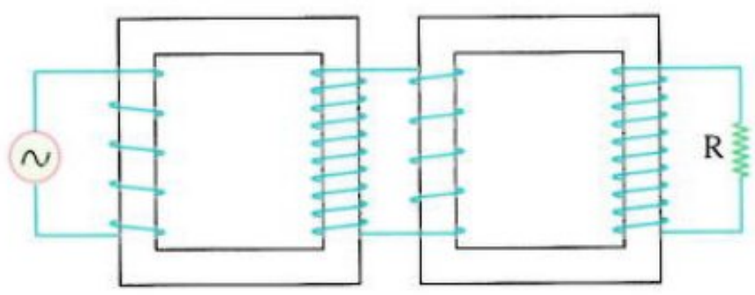
(a) 300 V, 150 V
(b) 120 V, 60 V
(c) 30 V, 50 V
(d) 30 V, 15 V

49 In the opposite figure, a non-ideal transformer that has two secondary coils. What are the probable values for the electric powers in these two coils?

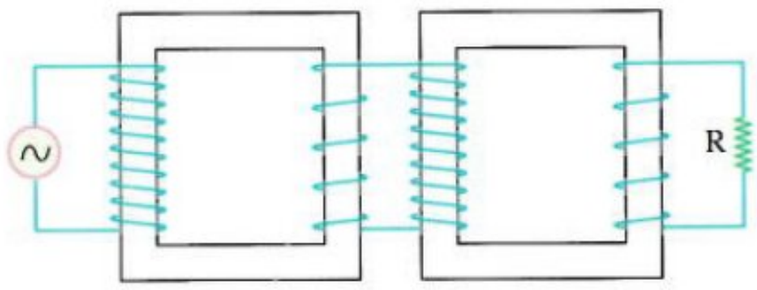


	$(P_w)_{s1}$	$(P_w)_{s2}$
(a)	120 W	120 W
(b)	100 W	100 W
(c)	80 W	160 W
(d)	120 W	60 W

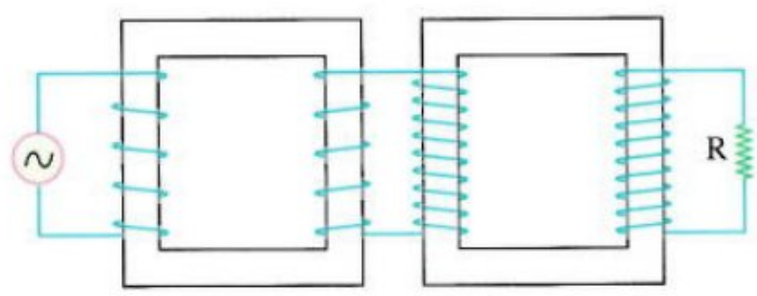
50 An AC source, a resistor (R) and four coils of turns 50, 50, 100, 100 were used four times to form two ideal transformers connected together each time as shown in the following figures, in which of these figures is the voltage difference across the resistor the greatest?



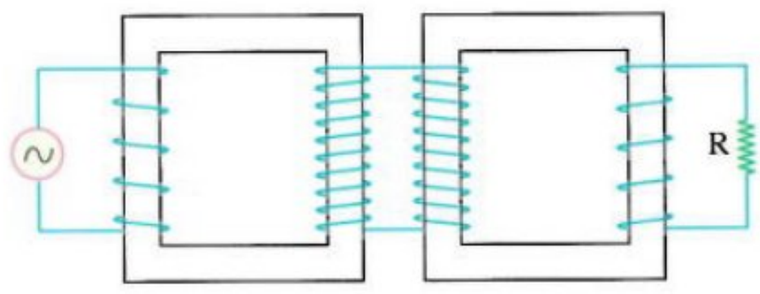
(a)



(b)



(c)

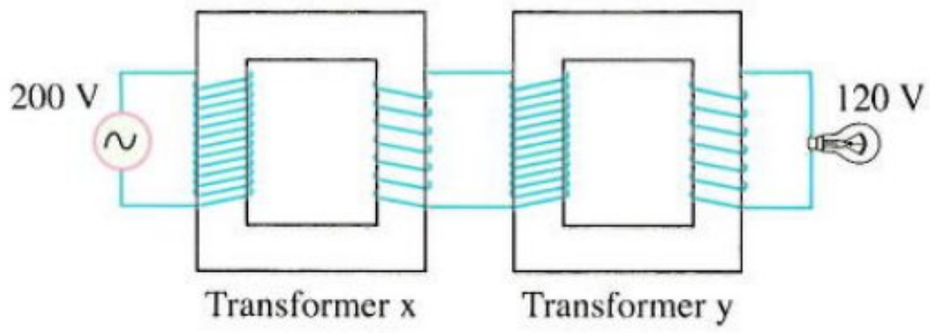


(d)

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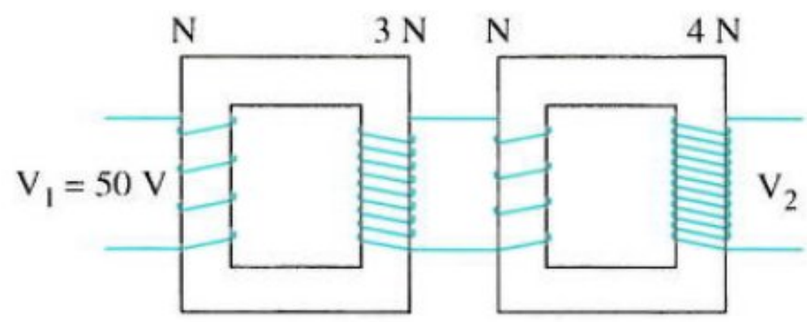
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51 In the opposite figure, two ideal electric transformers (x, y) are connected together. The primary coil of transformer x is connected to an AC source of 200 V and the secondary coil of transformer y is connected to an electric bulb that works on a potential difference of 120 V. If the ratio between the number of turns of the two coils of transformer x is $\frac{(N_s)_x}{(N_p)_x} = \frac{1}{3}$, then the ratio between the number of turns of the two coils of transformer y $\left(\frac{(N_p)_y}{(N_s)_y}\right)$ equals ..... .



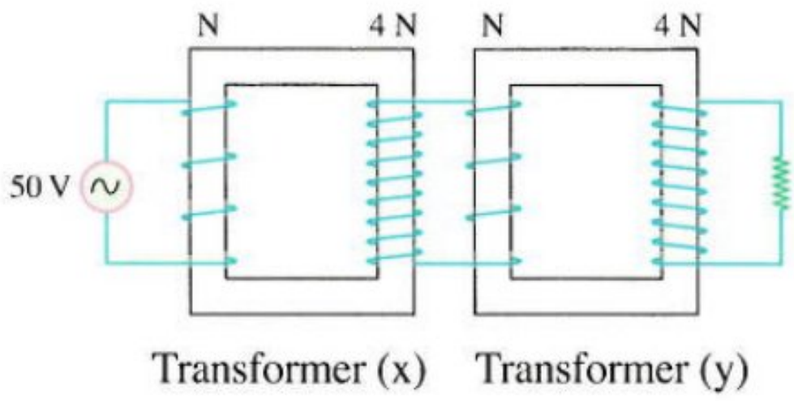
(a) $\frac{3}{8}$ (b) $\frac{2}{5}$ (c) $\frac{4}{7}$ (d) $\frac{5}{9}$

52 In the opposite figure, two ideal transformers are connected in series, then the value of V_2 equals ..... .



(a) 300 V (b) 450 V
(c) 600 V (d) 900 V

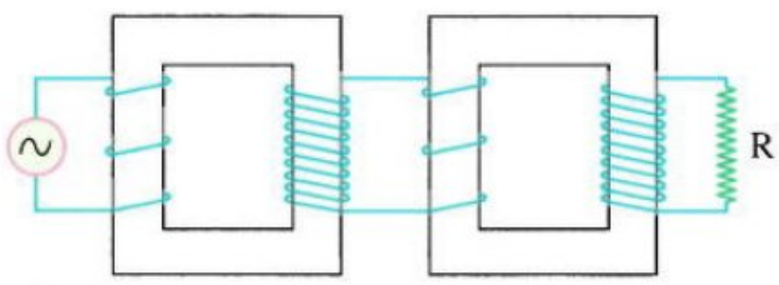
53 Two electric transformers (x , y) are connected together, the efficiency of each is 75% and the number of turns of their coils as shown in the opposite figure, if the voltage of the AC source is 50 V, then the voltage across the secondary coil of transformer (y) equals ..... .



(a) 150 V (b) 450 V (c) 600 V (d) 800 V

54 Two electric transformers are connected together as shown in the opposite figure , the efficiency of each is 90%, then

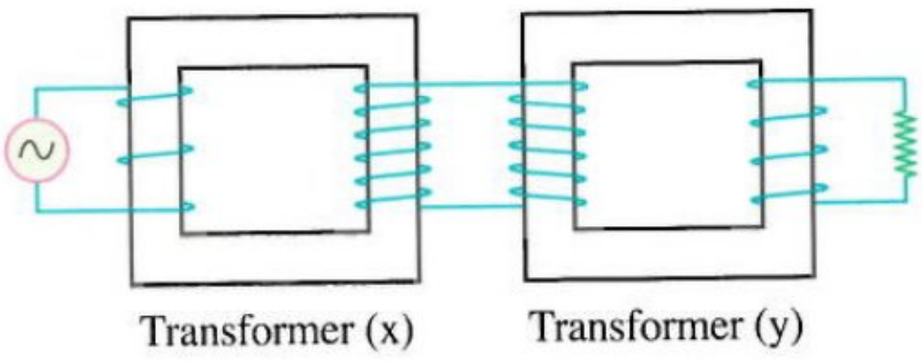
$\frac{\text{The electric power consumed in resistor R}}{\text{The electric power drawn from the source}}$ equals ..... .



(a) $\frac{9}{10}$ (b) $\frac{9}{100}$ (c) $\frac{81}{10}$ (d) $\frac{81}{100}$

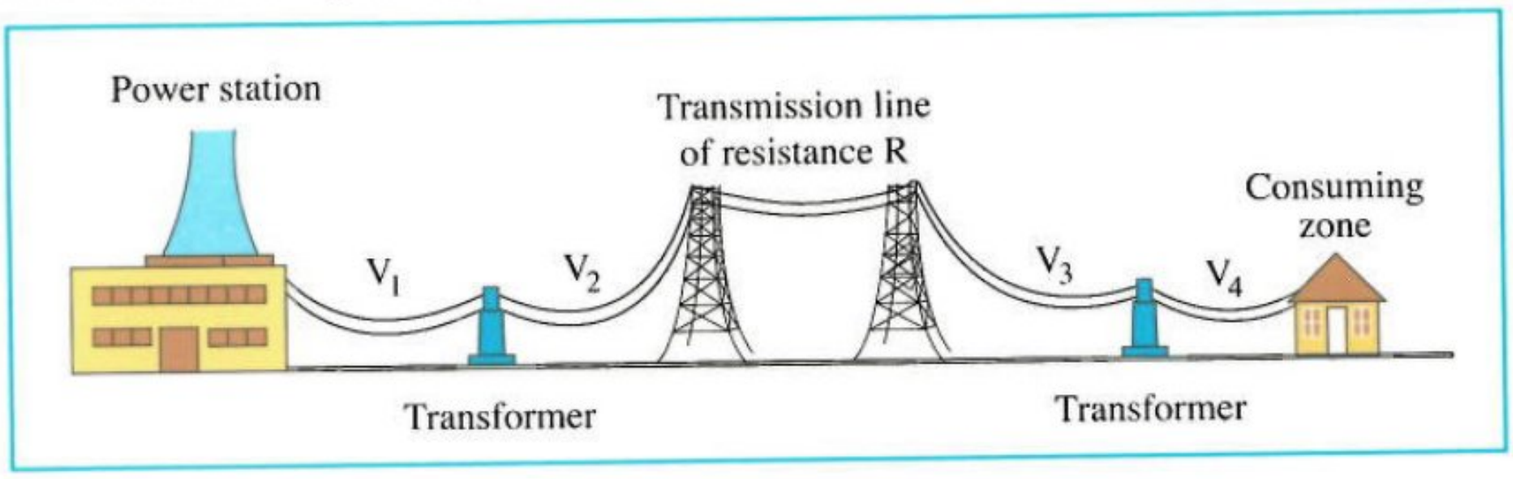
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55 ✱ In the opposite figure, two electric transformers (x, y) are connected together, the ratio of turns of the coils of transformer (x) is $\frac{1}{2}$ and it works on an AC source of voltage 220 V, while the ratio of turns of the coils in transformer (y) is $\frac{3}{2}$ and the voltage across the terminals of its secondary coil is 110 V, if the ratio between the efficiencies of the two transformers $\frac{\eta_x}{\eta_y}$ equals $\frac{3}{2}$, then the voltage across the terminals of the primary coil of transformer (y) equals ..... .



(a) 165 V (b) 330 V (c) 440 V (d) 660 V

56 The following figure represents the electric energy transmission from the electric power station to the regions of distribution,



which of the voltages shown on the figure is the highest ?

(a) V_1 (b) V_2 (c) V_3 (d) V_4

57 When transmitting the electric energy over long distances using electric transformers, if the voltage at the start of transmission lines is raised to ten times its original value, the loss in the consumed electric power in the transmission wires decreases to ..... of the consumed electric power that was supposed to be lost if the transformers weren't used.

(a) $\frac{1}{10}$ (b) $\frac{1}{100}$ (c) $\frac{1}{1000}$ (d) $\frac{1}{10000}$

58 ✱ An electric transformer, the ratio between the number of turns of its two coils is 1 : 5, is used to transfer an electric power of 100 kW with a potential difference of 200 volts from an electric power station to one of the places of its distribution. If the resistance of the transfer wires is 4 Ω , then the transfer efficiency is equal to ..... .

(a) 80% (b) 75% (c) 60% (d) 40%

59 ✱ An electric power of 200 kW is required to be transferred from the power station to one of the factories through a transmission line of resistance 0.5 Ω. If the potential difference at the power station is 1000 V, then:

(i) The electric current intensity in the transmission line equals ..... .

(a) 200 A (b) 20 A (c) 2×10^{-3} A (d) 2×10^{-4} A

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(ii) The drop in the voltage through the transmission line equals ..... .

(a) 10^{-4} V (b) 0.01 V (c) 10 V (d) 100 V

(iii) The lost power in the transmission line equals ..... .

(a) 2×10^{4} W (b) 200 W (c) 2×10^{-6} W (d) 2×10^{-8} W

60 ✻ Electric energy is transmitted from a power station through cables (wires) of total resistance 200 Ω. If the output electric voltage from station is 2×10^{4} V and the electric power produced from it is 400 kW, then the power lost in the cables as thermal energy equals ..... .

(a) 8×10^{4} W (b) 4×10^{3} W (c) 800 W (d) 2 W

61 ✻ An ideal electric transformer has a primary coil of N turns and two secondary coils x and y of 2 N, 4 N turns respectively, each secondary coil is connected to a resistor (R), if both secondary coils circuits are closed together, the current passing through the primary coil equals I_p, then the values of the currents through the two coils (x , y) respectively are ..... , ...

(a) $\frac{1}{4} I_p , \frac{1}{16} I_p$ (b) $\frac{1}{2} I_p , \frac{1}{8} I_p$ (c) $\frac{1}{3} I_p , \frac{2}{3} I_p$ (d) $\frac{1}{10} I_p , \frac{1}{5} I_p$

Electric motor

62 The device in which the electric energy is converted into mechanical energy is ..... .

(a) the electric transformer (b) the electric motor
(c) the electric generator (d) the electric bulb

63 The passing electric current in the coil of an electric motor during its rotation ..... .

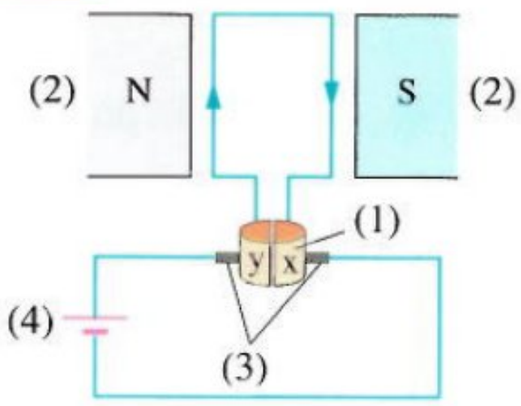
(a) has constant intensity and direction (b) has variable intensity and direction
(c) changes its direction every half cycle
(d) has an intensity which is directly proportional to the angle of rotation

64 In the electric motor; the current vanishes in the coil at the moment when ..... .

(a) the magnetic flux linked with the coil vanishes
(b) the magnetic flux density reaches its minimum value
(c) the magnetic torque which affects the coil vanishes
(d) the magnetic torque which affects the coil reaches half of its maximum value

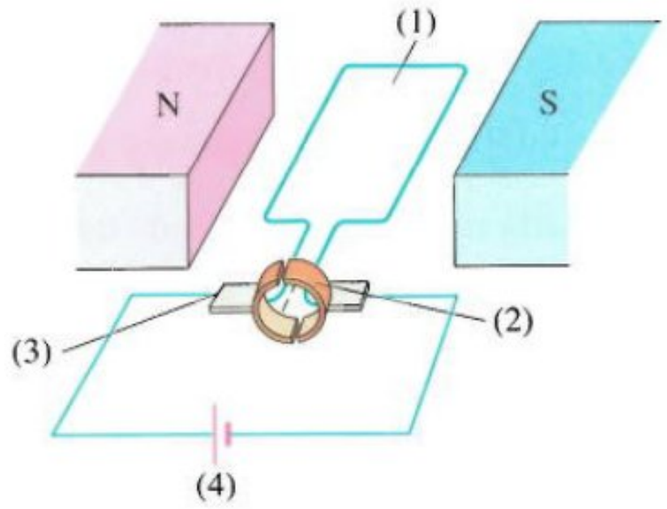
65 The opposite diagram shows a structure of a simple motor, which of the components in the diagram provides the motor with the required energy to rotate?

(a) Component 1 (b) Component 2
(c) Component 3 (d) Component 4



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66 The opposite figure represents the structure of a simple electric motor, which parts shown in the figure can be modified or replaced by another component so that the device:



(i) has a maximum mechanical power?

(a) The part (2) only
(b) The parts (1), (3) together
(c) The parts (1), (2) together
(d) The parts (2), (3) together

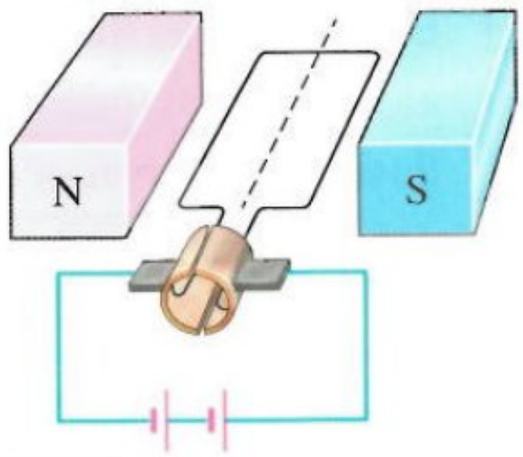
(ii) can be used to obtain a unidirectional electric current of variable intensity?

(a) Part (1) (b) Part (2) (c) Part (3) (d) Part (4)

(iii) can be used to obtain AC current?

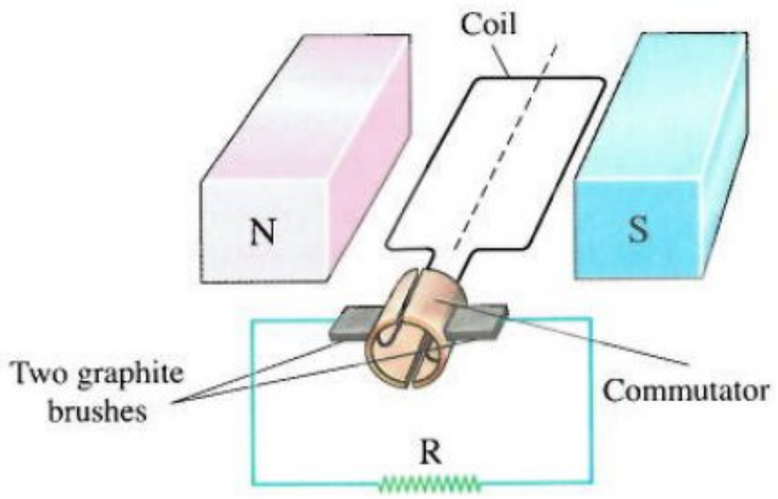
(a) The parts (1), (3) together (b) The parts (2), (4) together
(c) The parts (1), (2) together (d) The parts (2), (3) together

67 The opposite figure shows a simple motor, so the coil ..... .



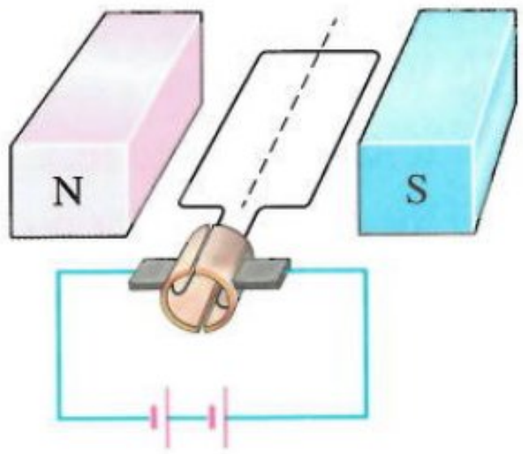
(a) rotates in the clockwise direction
(b) rotates in the counter-clockwise direction
(c) does not rotate
(d) the answer is indeterminable

68 The opposite diagram shows the structure of one of the electric devices. What happens if an electric cell is used instead of resistor R?



(a) The two brushes start to rotate.
(b) The commutator starts to rotate.
(c) The coil remains fixed and doesn't rotate.
(d) The current passes in the coil in one direction.

69 The opposite figure represents an electric motor whose coil is in a certain position, so the coil:



(i) moves from this position by the effect of .....

(a) the magnetic torque
(b) the inertia
(c) the electromagnetic induction
(d) the eddy currents

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(ii) after quarter a cycle, it continues its rotation from the shown position due to ..... .

(a) the magnetic torque (b) the inertia

(c) the electromagnetic induction (d) the eddy currents

(iii) its speed becomes uniform after a while of rotation because of ..... .

(a) the magnetic torque

(b) the inertia

(c) the eddy currents

(d) the generation of back induced emf in it

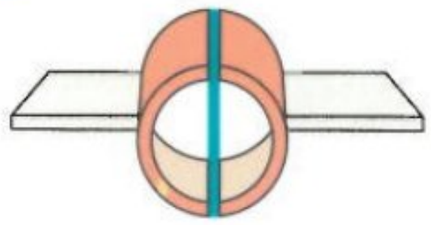
70 An electric motor has a coil resistance of 20 Ω connected to a DC source with an electromotive force of 220 V, neglecting the internal resistance, if the rotation speed of the motor coil stabilizes when the value of the induced back electromotive force in the coil is 180 V, then the value of the current passing through its coil after the rotation speed stabilizes is ..... .

(a) 2 A (b) 4 A (c) 20 A (d) 40 A

71 When an electric motor starts running, its speed gradually increases until it stabilizes at a certain value through an interval of time (t_o), what happens to both the induced back electromotive force in the coil and the value of the electric current passing through it during an interval of time (t_o)?

	The back induced emf in the coil	The value of the electric current in the coil
(a)	Increases until it stabilizes	Increases until it stabilizes
(b)	Increases until it stabilizes	Decreases until it stabilizes
(c)	Decreases until it stabilizes	Increases until it stabilizes
(d)	Decreases until it stabilizes	Decreases until it stabilizes

72 The opposite figure represents a position of the metallic commutator with respect to the graphite brushes in a motor, so the produced torque at this position is ..... .



(a) the maximum value (b) $\frac{1}{2}$ the maximum value

(c) $\frac{2}{3}$ the maximum value (d) zero

73 When a motor's coil starts its rotation from the moment at which its plane was parallel to the magnetic flux, then the value which decreases gradually until the coil reaches the perpendicular position is ..... .

(a) the torque that acts on the coil

(b) the magnetic forces that act on the two longitudinal sides of the coil

(c) the magnetic flux density affecting the coil

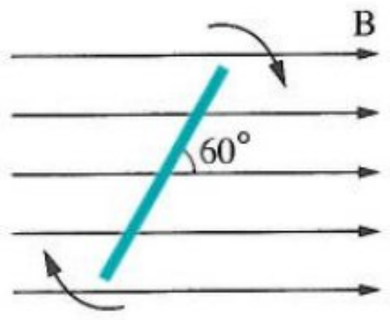
(d) the magnetic dipole moment of the coil

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74 During the rotation of the coil of a motor from the perpendicular position to the parallel position, the quantity that increases is ..... .

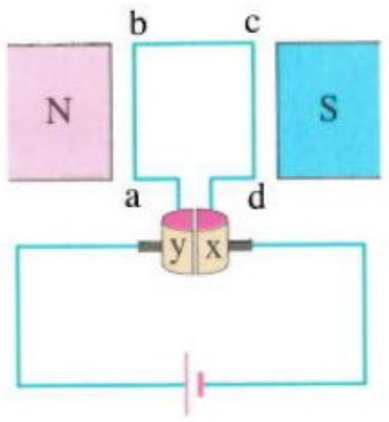
(a) the magnetic flux density that acts on the coil
(b) the magnetic flux that passes through the coil
(c) the force that acts on the two sides which are perpendicular on its axis
(d) the torque that acts on the coil

75 The opposite figure represents a motor's coil rotating from this position in clockwise direction, then the moment at which the current in the coil reverses its direction, will be after the coil rotates from this position by an angle of ..... .



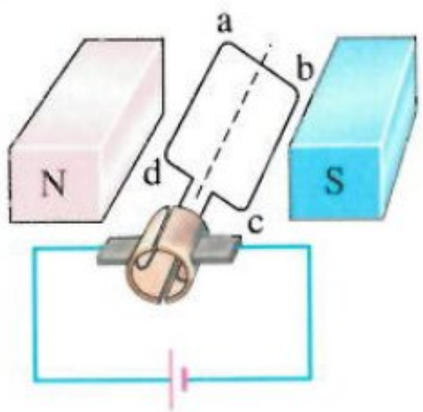
(a) 60°
(b) 90°
(c) 120°
(d) 150°

76 The opposite figure shows a square motor coil that rotates between the poles of a magnet, so the magnetic force acting on the side ab vanishes when the plane of the coil is ..... .



(a) perpendicular to the magnetic field
(b) parallel to the magnetic field
(c) inclined to the magnetic field by an angle 45°
(d) in all positions

77 In the opposite figure, an electric motor of dimensions $\ell, 2\ell$ inclined by an angle of 30° to the direction of the magnetic field, if the coil of the motor rotates with a uniform speed, so the ratio between the values of the magnetic force acting on the two sides ab, bc in the shown position in the figure $\left(\frac{F_{ab}}{F_{bc}}\right)$ is ..... .



(a) $\frac{1}{2}$
(b) $\frac{1}{4}$
(c) $\frac{\sqrt{3}}{2}$
(d) $\frac{1}{1}$

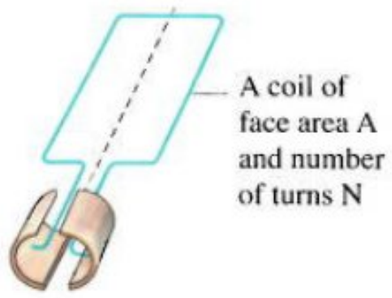
78 What happens to the maximum speed of rotation at which the coil of an electric motor rotates when increasing the electromotive force of the source?

(a) Increases.
(b) Decreases.
(c) Remains unchanged.
(d) The answer can't be determined.

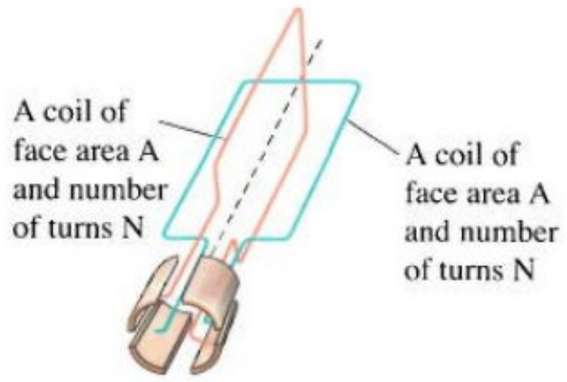
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79 Which of the two designs (1) and (2) shown in the two opposite figures for a coil of an electric motor can provide greater mechanical power when connected to the same DC source and placed in the same magnetic field?



Design (1)



Design (2)

(a) Design (1). (b) Design (2).
(c) Both designs provides the same power.
(d) Neither design is suitable as a coil for the motor.

80 Figure (1) shows the structure of an electric motor connected to a battery, then modifications were made to the motor's design by placing a soft iron cylinder that is divided into isolated discs at the core of the coil and using a concave magnet poles as shown in figure (2), what is the effect of this on the motor's efficiency? And why?

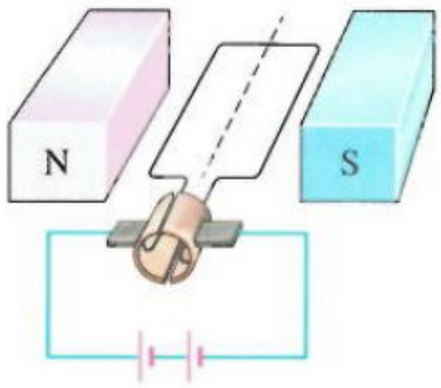


Figure (1)

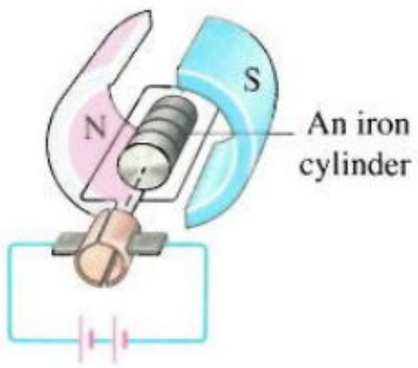


Figure (2)

(a) The efficiency increases, because the coil at any position is perpendicular to the direction of the field.
(b) The efficiency increases, because the coil at any position is parallel to the direction of the field.
(c) The efficiency decreases, because the coil at any position is perpendicular to the direction of the field.
(d) The efficiency decreases, because the coil at any position is parallel to the direction of the field.

Second Essay questions

Electric transformer

1 **Clarify with two examples** how choosing specific materials to make the components of an electrical transformer increases its efficiency.

2 **In the electrical transformer, explain the reason for each of the following:**
(a) The difference in the number of turns of two coils.
(b) Using an alternating source.

3 The mobile charger contains a step-down transformer, **why does** its temperature increase while working and **how does** that affect the efficiency of the transformer?

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4 In most coils used for AC currents, the cores of the coils are made of electrically insulated parallel thin sheets of soft iron, instead of being one piece of iron. **Show the practical benefit of this**.

5 An electric transformer has efficiency of 80%, its secondary coil has a smaller number of turns of a thicker wire than that of its primary coil:

(a) **Is** the transformer a step-up or a step-down transformer?

(b) **Why** are the turns of the secondary coil made from a wire thicker than that of the primary coil?

6 **Explain how** the current through the primary coil of an electrical transformer cause the generation of an induced emf in the secondary coil.

7 **Show why** the step-down ideal transformer raises the current and the step-up ideal transformer reduces the current.

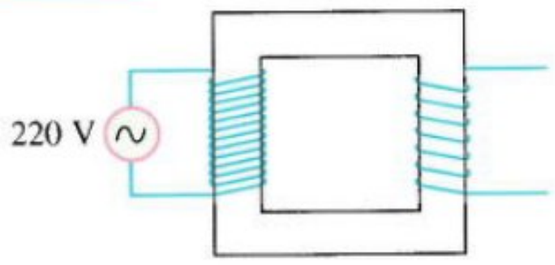
8 In the step-up transformer, the potential difference across the terminals of the secondary coil is greater than that across the terminals of the primary coil, **does** this violate the law of conservation of energy? **Explain your answer**.

9 An ideal electric transformer, the number of turns of its coils are 800, 400 turns, is connected to an AC source of emf 100 V, **calculate** the maximum and the minimum emf that can be obtained using this transformer.

10 **Find the ratio** of the potential difference across one turn of the primary coil to that of the secondary coil $\frac{(V_p)_{turn}}{(V_s)_{turn}}$, if the efficiency of the transformer is:

(a) 100% (b) 90%

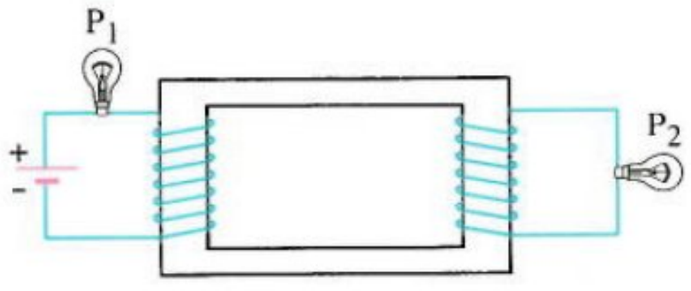
11 In the opposite figure, a primary coil of an ideal transformer is connected to an AC dynamo of voltage 220 V and frequency 50 Hz:



(a) **What** are the characteristics of the induced voltage across the terminals of the secondary coil?

(b) **Explain what happens to** this voltage when increasing the frequency of rotation of the dynamo coil.

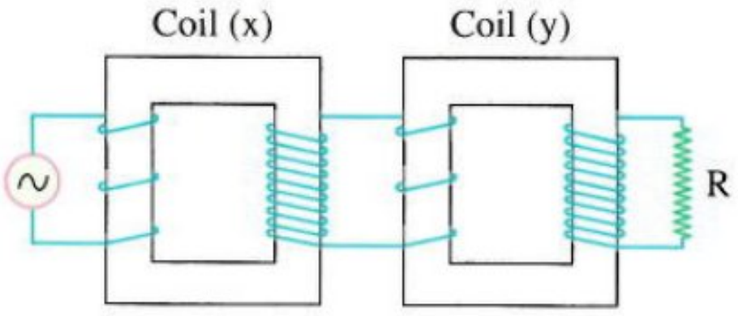
12 In the opposite figure: An electric transformer has two coils of equal numbers of turns and its primary coil is connected to a DC source, **which** of the two lamps P_1, P_2 glows? **Explain your answer**.



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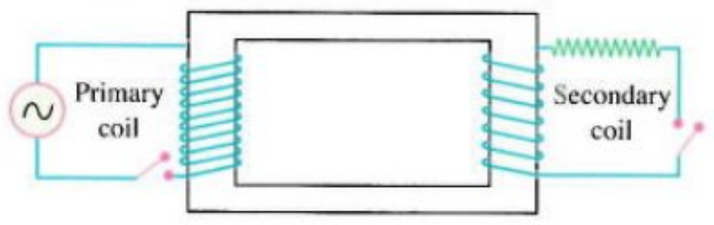
13 The opposite figure shows two ideal electrical transformers x, y, the ratio of turns of the coils of each of them is $\left(\frac{N_p}{N_s} = \frac{1}{3}\right)$, **find the ratio between:**



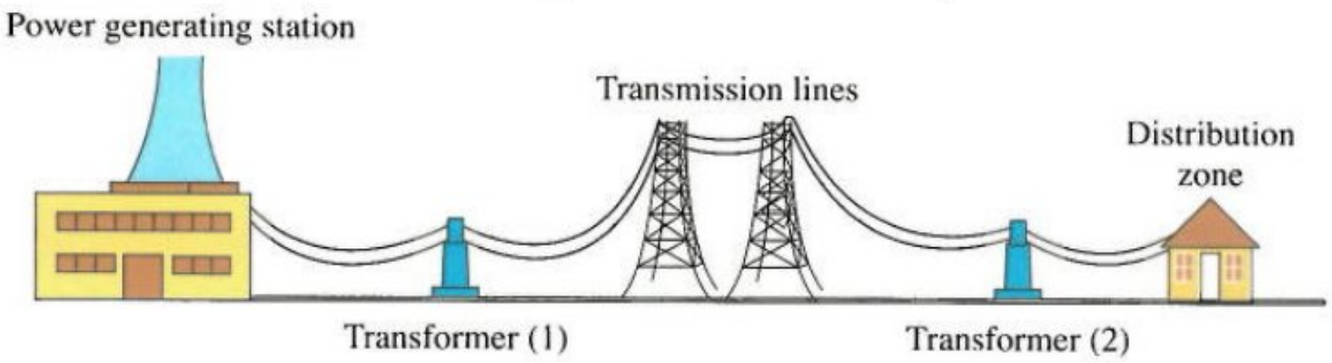
(a) The power consumed in resistor R and the power drawn from the source.

(b) The voltage across the terminals of the secondary coil of transformer (y) and the voltage across the terminals of the primary coil of transformer (x).

14 The opposite figure shows an electric transformer, **explain what happens when** closing the circuit of the primary coil and opening the circuit of the secondary coil concerning the energy consumed in the primary coil.



15 The following figure illustrates the long distances electric power transmission,



(a) **Give reason for:** Using two different types of electrical transformers in the electric power transmission.

(b) **What are the expected consequences on the efficiency of transmission assuming:**

① The places of the two transformers (1), (2) are exchanged with each other?

② Not using any electric transformers in the long-distance power transmission?

Electric motor

16 **Why does** the rotation of the electric motor coil continue when it passes by the perpendicular position to the magnetic field?

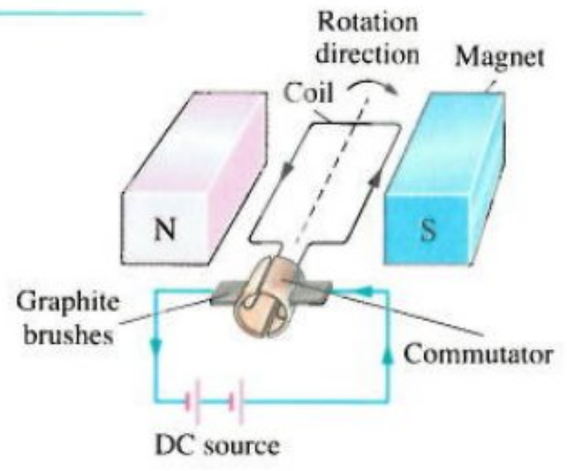
17 **Why does** the speed of rotation of the electric motor coil stabilize after some time of operating it?

18 **How can** the power of the electric motor be increased?

19 **The opposite figure illustrates the structure of an electric motor:**

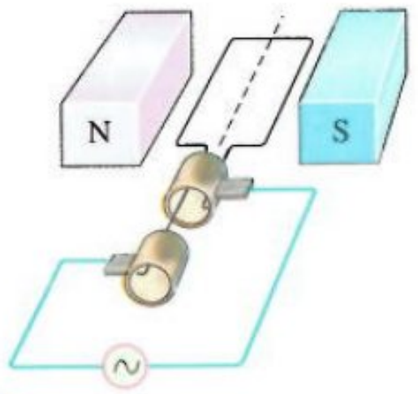
(a) **Why does** the motor coil rotate in one direction of rotation?

(b) **How can** the direction of rotation of the motor coil be reversed?



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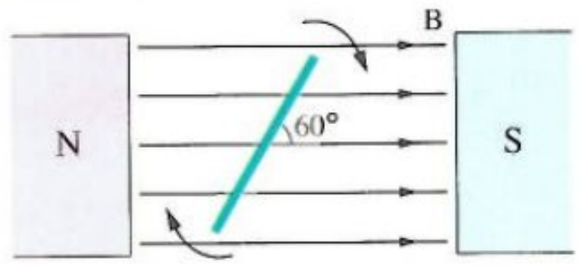
20 The opposite figure illustrates a proposed prototype of an electric motor, **mention two modifications** that should be done on this prototype to make its coil rotate in only one direction.



21 **Explain:** The current direction through the coil of the electric motor gets reversed despite connecting it to a DC source.

22 The opposite figure illustrates a coil of electric motor rotating with a constant speed of periodic time T in a uniform magnetic field, **what is** the least time interval, in terms of T, taken by the coil to rotate from the shown position till the position at which the torque affecting the coil equals:

(a) zero? (b) its maximum value?



23 **Explain:** The coil of the electric motor gets spoiled if it is prevented to rotate while it is connected to a DC source.

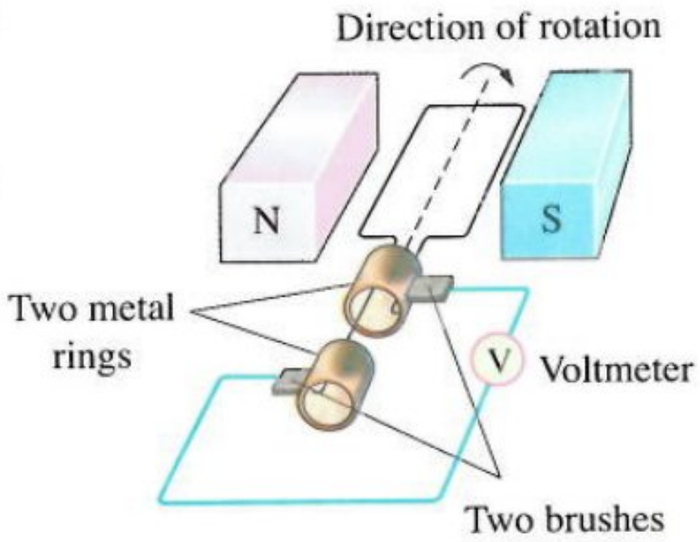
24 **Compare between** the dynamo and the motor (in terms of: the role of the cylinder commutator).

25 **Compare between** the reasons for the presence of multiple coils between which small equal angles in each of the DC dynamo and the electric motor.

26 The opposite figure represents a simple dynamo where a student wants to convert it into a motor to rotate a small fan, so he replaced the voltmeter with a battery and a switch and when he closed the switch the motor could not rotate the fan as in the ordinary motors:

(a) **What** is the reason in the structure of the motor preventing its coil to rotate as usual?

(b) **How** can you help the student to make the fan rotate as in the normal electric motors? **Illustrate by drawing.**



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Questions of the Previous Years Exams on Lesson 4

The electric transformer

1 An electric bell works at a power of 1 W when an electric current of 0.5 A passes through it. If it is connected to a transformer of efficiency 95% in which the number of turns in its secondary coil is $\frac{1}{100}$ of that in its primary coil, then the voltage of the source connected to the primary coil equals ..... . *(Exp. May 21)*

(a) 105.26 V (b) 110.34 V (c) 210.53 V (d) 215.62 V

2 The ratio between the number of turns of the coils of an ideal step-up transformer is $\frac{3}{2}$. If its secondary coil is connected to a device that works on a potential difference of 300 V, so the values of V_P and $\left(\frac{(P_w)_s}{(P_w)_p}\right)$ respectively are ..... , ... *(Exp. June 21)*

(a) 200 V, $\frac{2}{3}$ (b) 450 V, $\frac{3}{2}$ (c) 200 V, $\frac{1}{1}$ (d) 450 V, $\frac{1}{1}$

3 In an ideal step-down transformer, the ratio between the number of its turns is $\frac{4}{1}$, so if its secondary coil is connected to an electric bulb labelled (60 V, 20 A), so the choices which express the values of (I_p) and (V_p) correctly when the bulb works with its full power respectively are ..... , ... *(1st session 21)*

(a) 40 A, 150 V (b) 5 A, 240 V (c) 80 A, 240 V (d) 5 A, 15 V

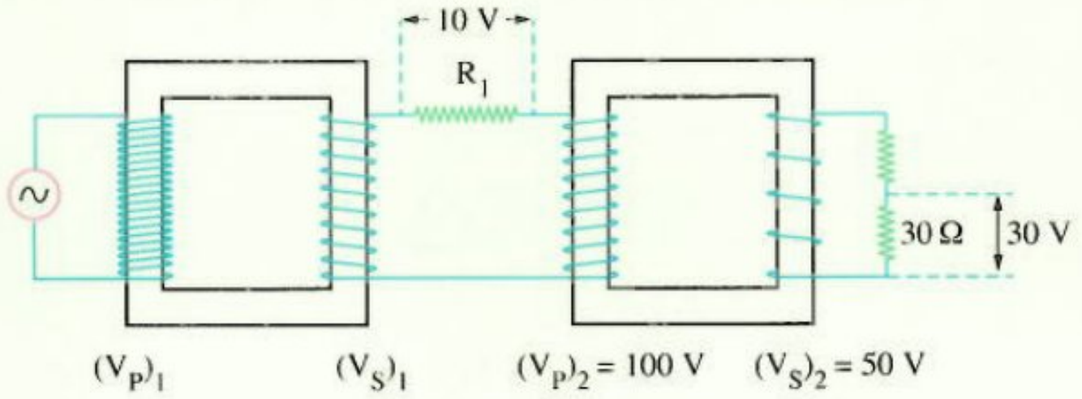
4 A step-down transformer has efficiency of 90%, the ratio between potential difference of its two coils is $\frac{4}{7}$ and the electric current intensity passing in the primary coil is 10 A. If the number of turns of the primary coil is 400 turns, so the choice which correctly expresses the values of (I_s) and (N_s) is ..... . *(2nd session 21)*

(a) 15.75 A, 229 turns (b) 17.5 A, 229 turns

(c) 15.75 A, 254 turns (d) 17.5 A, 254 turns

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5 Two ideal electric transformers are connected together as shown in the figure.

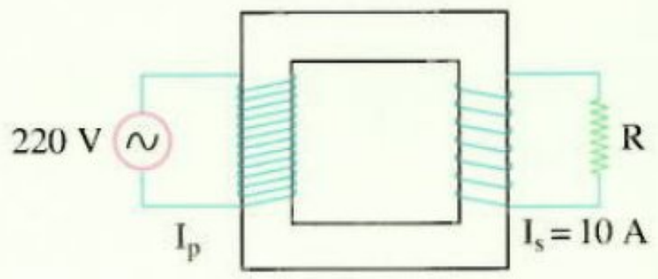


Using the shown data in the figure, the consumed electric power in the resistance (R_1) equals ..... .

(1st session 22)

(a) 10 watts (b) 50 watts (c) 55 watts (d) 5 watts

6 The figure shows a step-down electric transformer with efficiency of 80% and the ratio between the number of turns of its two coils is $\frac{3}{5}$, then the value of the produced voltage at the secondary coil equals to ..... while the current intensity through the primary coil equals to ..... .



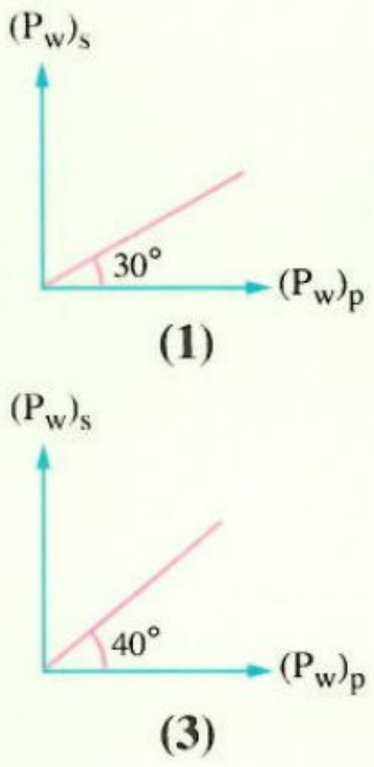
(2nd session 22)

(a) 108.3 V, 6 A (b) 105.6 V, 8 A (c) 108.3 V, 8 A (d) 105.6 V, 6 A

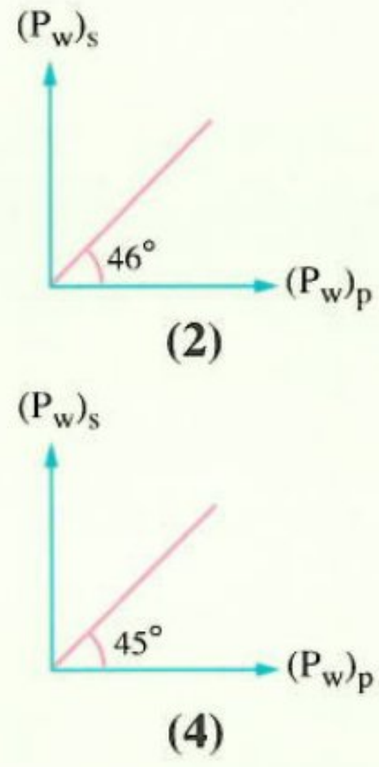
7 A step-down transformer of efficiency 90% is used to operate a bell written on it (60 W - 0.5 A), if the transformer works by 220 V, so the ratio $\left(\frac{N_s}{N_p}\right)$ = ... *(1st session 24)*

(a) $\frac{33}{20}$ (b) $\frac{6}{11}$ (c) $\frac{11}{6}$ (d) $\frac{20}{33}$

8 Which of the following graphs represents a transformer having the highest efficiency? *(1st session 24)*



(1)



(2)

(3)

(4)

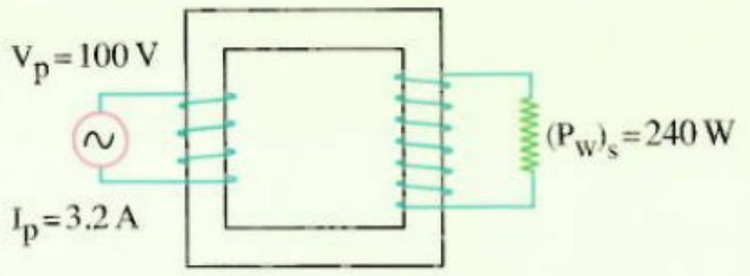
(a) Figure (4) (b) Figure (2) (c) Figure (1) (d) Figure (3)

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9 From the data shown in the opposite figure then: *(2nd session 24)*

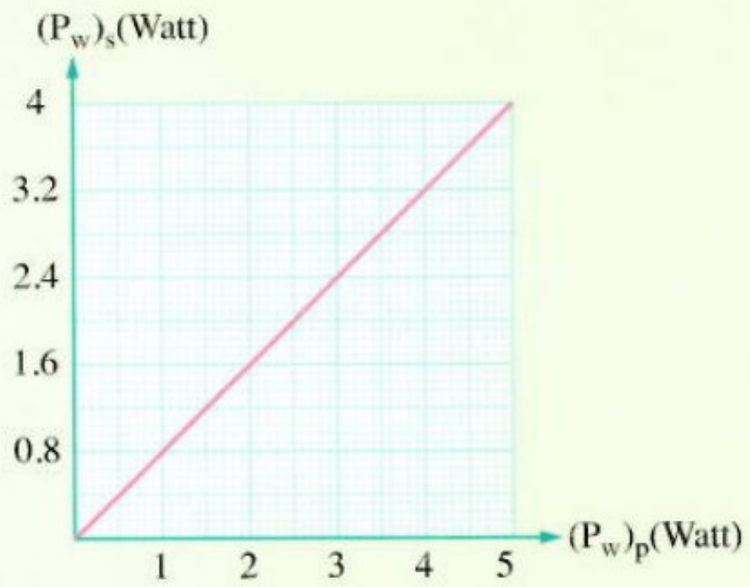


	Transformer efficiency	Type of transformer
(a)	100%	Step-up
(b)	100%	Step-down
(c)	75%	Step-up
(d)	75%	Step-down

10 Electric transformer of efficiency 90% is connected to alternating current source of power 60 kW, then the power produced from the secondary coil is ..... . *(2nd session 24)*

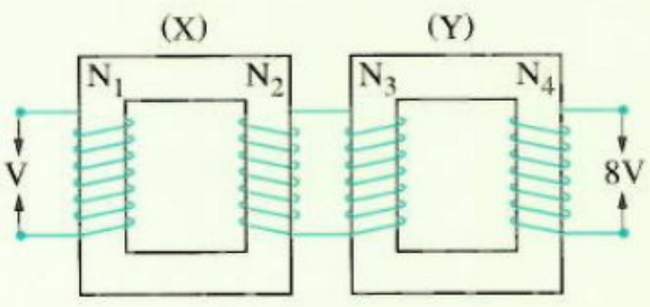
(a) 54 kW (b) 60 kW (c) 45 kW (d) 66.66 kW

11 The opposite graph represents the relation between the output power $(P_w)_s$ of an electric transformer and its input power $(P_w)_p$, if the voltage applied to the primary coil is 15 V, the induced voltage across the secondary coil is 72 V and the current passing through the secondary coil is 2 A, then the value of the current passing through the primary coil equals ..... . *(Practice 25)*



(a) 2.5 A (b) 7.68 A
(c) 9.6 A (d) 12 A

12 The figure represents two ideal transformers (X, Y), if the output voltage is equal to eight times the input voltage , then the number of turns in each transformer could be ..... . (Note that: the number of turns shown in the figure does not represent the actual values for the two transformers) *(Practice 25)*



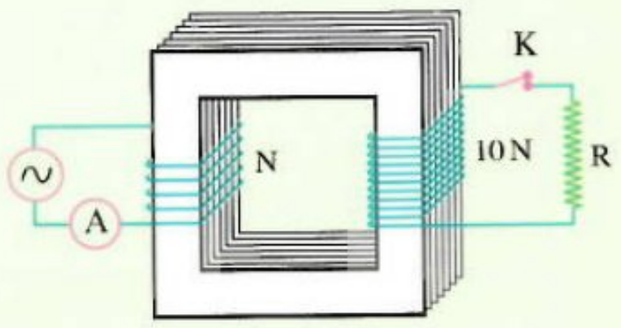
(a) $N_2 = 2\,N_1, N_4 = 4\,N_3$
(b) $N_2 = 2\,N_1, N_4 = 3\,N_3$
(c) $N_1 = 3\,N_2, N_3 = 2\,N_4$
(d) $N_1 = 2\,N_2, N_3 = 4\,N_4$

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13 An electric transformer raises the voltage from 120 V to 10^3 V and reduces the current from 10^3 A to 110 A, the lost electric power is equal to ..... . *(Practice 25)*

(a) 5×10^3 W (b) 5×10^4 W (c) 10^3 W (d) 10^4 W

14 The opposite figure represents an ideal electric transformer, if the reading of the hot-wire ammeter when the switch is closed is 1 A, then its reading after opening the switch ..... . *(Practice 25)*



(a) remains 1 A (b) becomes 10 A
(c) becomes 5 A (d) approaches zero

15 The following choices are related to the electrical conductivity of the materials of the transformer core and the wires of its coils, which choice can greatly improve the efficiency of the electric transformer? *(1st session 25)*

	Core material	Coil wire material
(a)	High conductivity	High conductivity
(b)	High conductivity	Low conductivity
(c)	Low conductivity	High conductivity
(d)	Low conductivity	Low conductivity

16 Which statement describes the characteristics of the transformer used at the regions where the electric energy is distributed? *(2nd session 25)*

(a) The current value in the primary coil is less than the current value in the secondary coil.
(b) The voltage across the primary coil is lower than the voltage across the secondary coil.
(c) The number of turns in the primary coil is lower than the number of turns in the secondary coil.
(d) The electric power in the primary coil is lower than the electric power in the secondary coil.

17 Step-down transformer of efficiency 100% operates on AC voltage of 220 V. It has two secondary coils (X) and (Y). Coil (X) is connected to an electric bell, and coil (Y) is connected to a lamp. If the bell and lamp are operating simultaneously, the potential difference across the bell is 20 V, with a current of 1A, and the potential difference across the lamp is 19 V, with a current of 0.8 A, what is the current flowing through the primary coil? *(2nd session 25)*

(a) 0.09 A (b) 0.2 A (c) 0.11 A (d) 0.16 A

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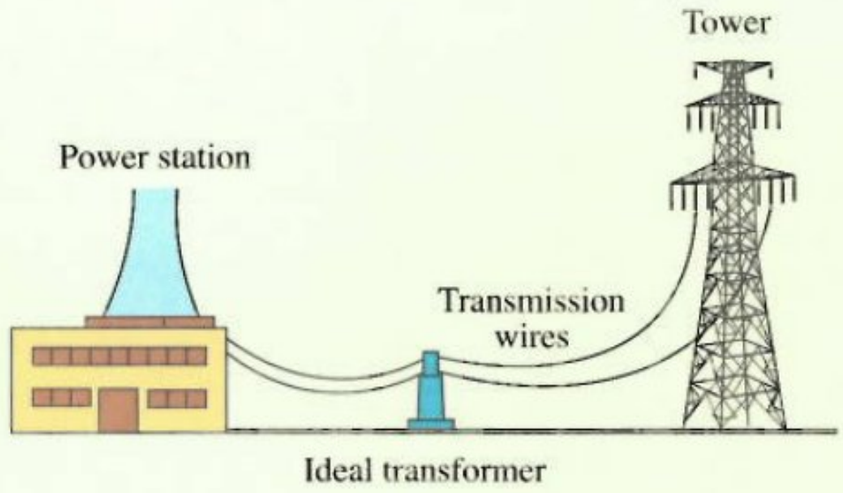
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18 Ideal electric transformer, its primary coil is connected to AC supply of 120 V and its secondary coil is connected to an electric lamp which works on 12 V and power 60 watts. **Calculate** the current intensity in both of the primary and secondary coil of the transformer. *(Exp. 23)*

.....

.....

19 In the opposite figure, through transferring electric energy from a power station at which the potential difference equals 25×10^3 V using an ideal transformer, the potential difference at one of the transition towers was 132×10^3 V, if the cables resistance between the tower and the transformer is 7500 Ω and the value for the current passing in them is 2 A, **calculate:**



(a) The potential difference between the two terminals of the secondary coil of the transformer.

(b) The current passing in the primary coil of the transformer. *(1st session 23)*

.....

.....

20 An ideal electric transformer, the number of turns of its primary coil is 1100 turns and works on a voltage of 220 V, if it has two secondary coils, one of them is connected to an electric bell labelled (6 V, 0.4 A) and the other is connected to an electric bulb labelled (12 V, 0.35 A), **then find:**

(a) The number of turns of the two secondary coils.

(b) The value of the primary coil current when both the bell and the bulb are operating together. *(Practice 25)*

.....

.....

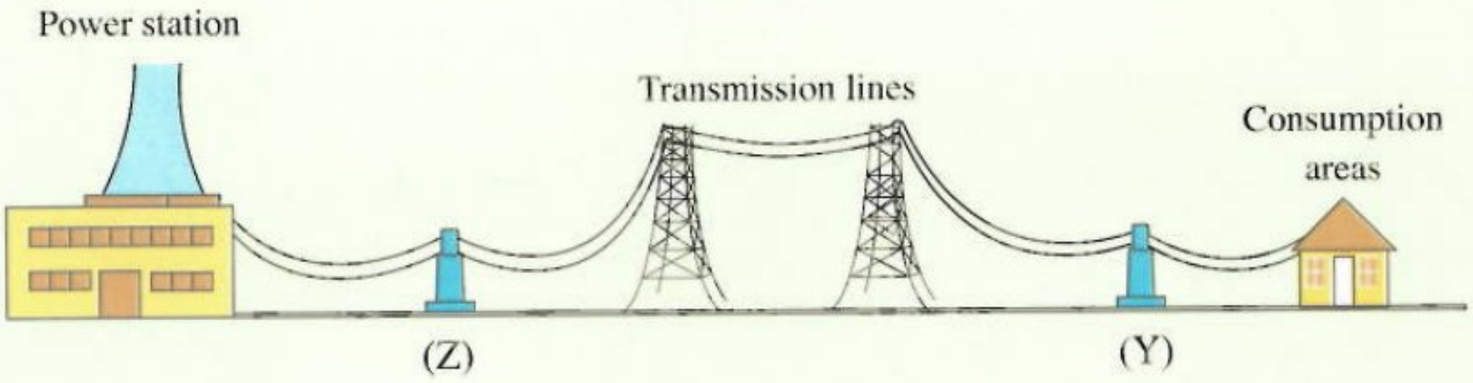
21 A power station generates an electric power of 100 kW at a voltage of 200 V, which is connected to an ideal electric transformer that the ratio between the number of turns of its coils is 1 : 5, **calculate** the transmission efficiency of a wire of resistance 4 Ω that is used to transfer this power. *(Practice 25)*

.....

.....

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22 The diagram illustrates the process of transferring electrical energy from a power plant to the consumption areas.



(a) What is the type of transformer (Y) and transformer (Z) used in this process?

(b) Explain how the efficiency of the energy transmission process can be improved by controlling two physical quantities related to transmission lines. *(1st session 25)*

.....

.....

.....

Electric motor

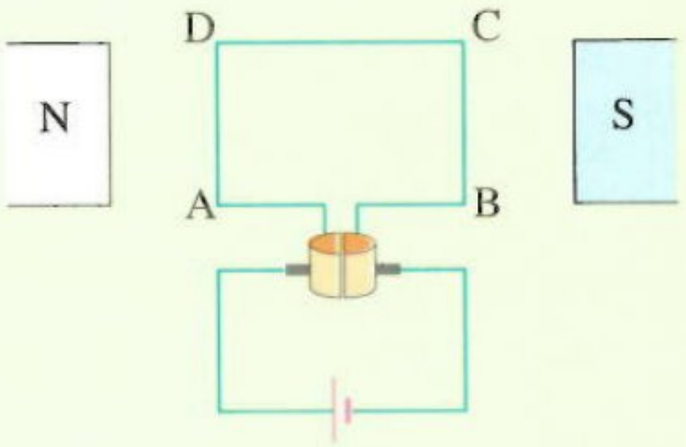
23 The figure illustrates the structure of a simple electric motor. When the coil rotates by an angle of 45° from the parallel position, the force acting on wire AD ..... . *(Exp. June 21)*

(a) remains maximum

(b) remains zero

(c) increases from zero to the maximum value

(d) decreases from the maximum value to zero



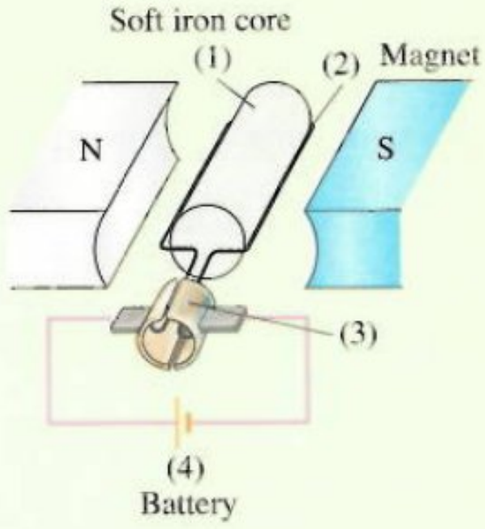
24 The opposite figure illustrates the structure of a simple electric motor. To minimize the eddy currents in the soft iron core, we must ..... .

(a) replace part number (3) with two metallic rings

(b) replace part number (1) with an iron core that is formed of thin insulated sheets

(c) replace part number (4) with a battery of larger emf

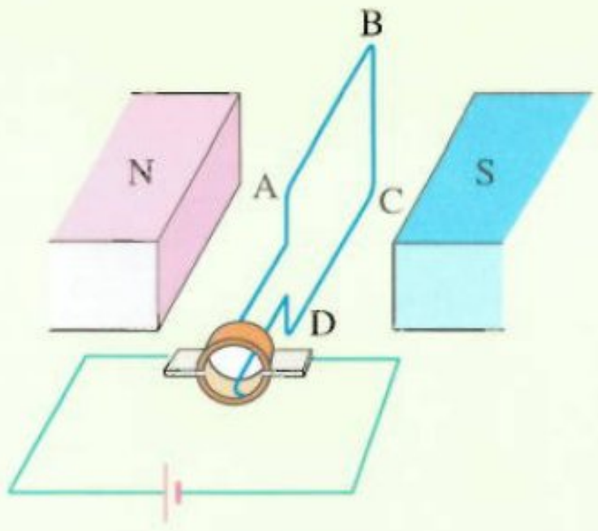
(d) replace part number (2) with a number of coils that have small angles between them *(1st session 21)*



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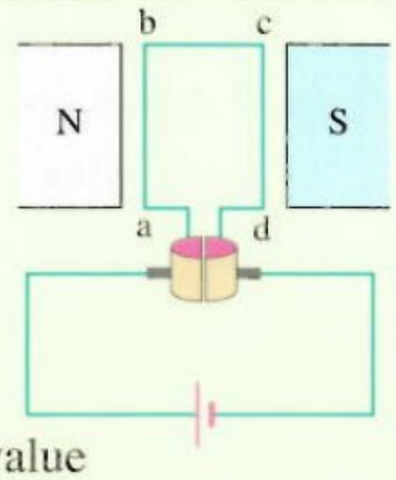
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25 The figure illustrates the structure of a simple electric motor. The coil ABCD continues rotation at the vertical position due to ..... . *(2nd session 21)*



(a) the force acting on wire (AB)
(b) the force acting on wire (BC)
(c) the inertia of the coil
(d) the force acting on the coil

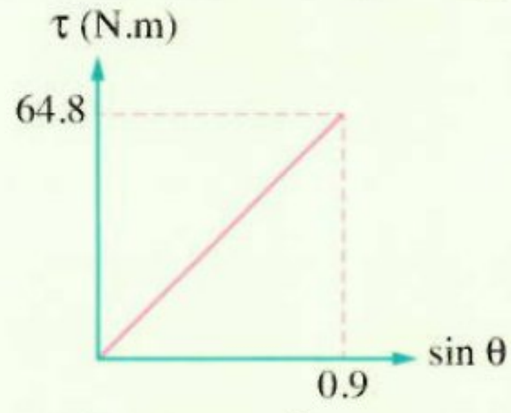
26 You have a DC electric motor consists of a single coil that started its motion from a position parallel to the flux lines as in the figure, after the coil rotates by 60° clockwise from the shown position, therefore ..... .



(a) the magnetic torque remains constant
(b) the magnetic force acting on side (bc) becomes half its maximum value
(c) the magnetic torque becomes $\frac{\sqrt{3}}{2}$ from the maximum value
(d) the magnetic force acting on the side (ab) remains constant *(1st session 23)*

27 A single coil DC electric motor, when the coil's plane is perpendicular to the magnetic flux lines, which of the following quantities **does not** equal zero?
(a) Magnetic dipole moment of the coil.
(b) Speed of rotation.
(c) Magnetic torque affecting the coil.
(d) The magnetic force affecting the coil's sides. *(2nd session 23)*

28 The opposite graph represents the relation between the value of the magnetic torque (τ) acting on the coil of an electric motor and the sine (sin θ) of the angle between the normal to the coil's plane and the magnetic flux lines, if the magnetic flux density affecting the coil is 0.3 T, then the magnetic dipole moment acting on the coil is equal to ..... . *(Practice 25)*



(a) 180 $A.m^2$ (b) 200 $A.m^2$ (c) 240 $A.m^2$ (d) 280 $A.m^2$

29 The role of the cylinder split into two isolated halves in the operation of the electric motor is ..... . *(Practice 25)*
(a) converting of DC current to AC current
(b) converting of AC current to DC current
(c) reversing the direction of the current in the coil every half a cycle
(d) generating of a unidirectional current

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30 Increasing the number of coils and longitudinally dividing the split cylinder into a number of parts equal to twice the number of coils in the electric motor is ..... . *(Practice 25)*

(a) to make the coil rotates in one direction

(b) to maintain a constant torque at the maximum value

(c) to keep the coil rotating

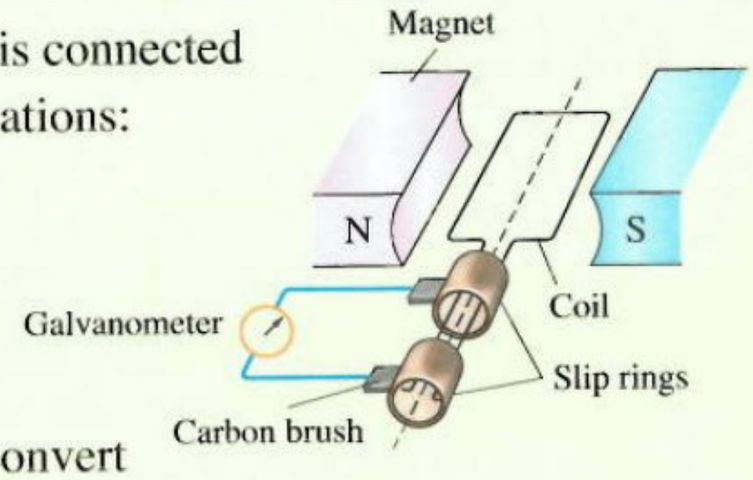
(d) to make the coil rotate at an increasing speed

31 The diagram shows an electric generator whose coil is connected to a galvanometer. You have three suggested modifications:

(I) Replacing the galvanometer with an AC supply.

(II) Replacing the galvanometer with a battery.

(III) Replacing the two slip rings with a metal cylinder split into two halves.



Which of the above modifications is/are required to convert the generator shown in the figure into an electric motor?

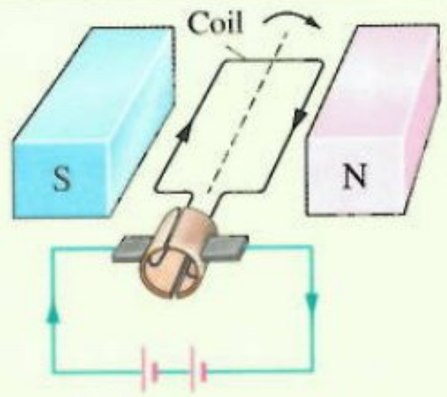
(a) Modification (II) only.

(b) Modification (I) only.

(c) Both modifications (II) and (III).

(d) Both modifications (I) and (III).

32 The diagram shows a certain position of the coil of an electric motor between the magnetic poles. What physical property is responsible for keeping the coil rotation continues after rotating 90° from the position shown in the diagram? *(2nd session 25)*



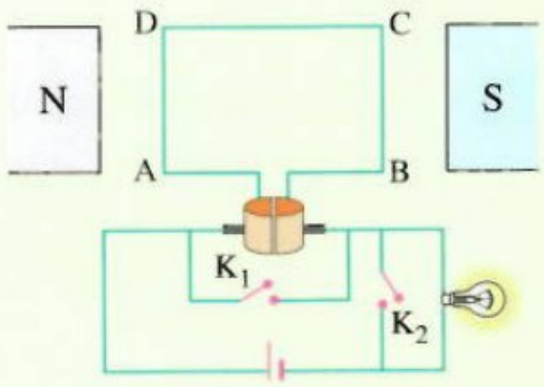
(a) Magnetic torque.

(b) Electromagnetic induction.

(c) Inertia.

(d) Eddy currents.

33 In the opposite figure, an electric motor has a coil of resistance R that starts rotation from the parallel position to a magnetic field. The coil is connected to a battery, an electric bulb and two switches, K_1, K_2, which of the following choices is correct when closing the switches K_1, K_2 separately? *(Practice 26)*



	When closing the switch K_1 only	When closing the switch K_2 only
(a)	The bulb goes off and the motor speed decreases	The motor stops and the luminosity of the bulb increases.
(b)	The bulb goes off and the motor speed decreases	The bulb goes off and the motor speed increases
(c)	The motor stops and the luminosity of the bulb increases	The motor stops and the luminosity of the bulb increases
(d)	The motor stops and the luminosity of the bulb increases	The bulb goes off and the motor speed increases

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Test on Chapter 3

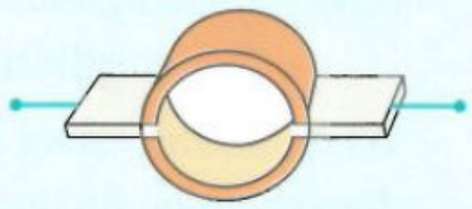
Answered

First Choose the correct answer

1 A wire of length 1 m moves in a uniform magnetic field of flux density 0.2 T with a uniform velocity of 1 m/s in a direction perpendicular to its length, so an induced emf of 0.1 V is generated across its ends, then the inclination angle of the velocity of the wire to the direction of the magnetic field is ..... .

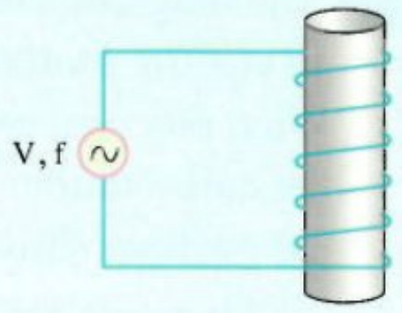
(a) 0° (b) 30° (c) 60° (d) 90°

2 The opposite figure shows one of the positions of the metallic commutator with respect to the graphite brushes in a motor during the rotation, so that the motor continues its rotation and skips this position because of ..... .



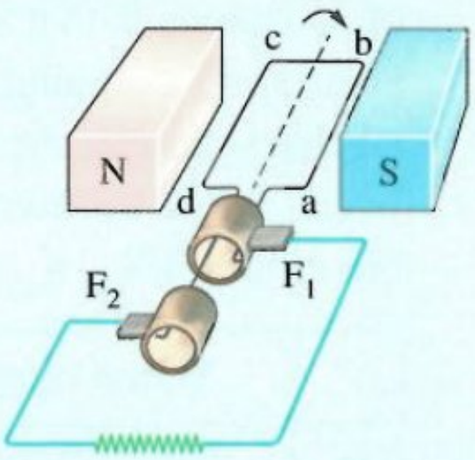
(a) the magnetic torque (b) the back induced emf
(c) the emf of the source (d) the inertia

3 The opposite figure represents a solid metal cylinder that is used as a core for a solenoid of an insulated wire which is connected to an AC dynamo that its effective voltage is (V) and its frequency is (f), what happens to the value of the generated eddy currents in the cylinder when increasing the dynamo frequency?



(a) Increases. (b) Decreases but does not vanish.
(c) Decreases. (d) Remains unchanged.

4 A dynamo coil abcd rotates with a periodic time T in the clockwise direction, so after a time interval of 2.25 T from the position shown in the opposite figure, the potential of brush F_1 becomes ..... the potential of brush F_2?



(a) greater than
(b) less than
(c) equals
(d) the answer is indeterminable

5 In AC dynamo, if the time required by the current to get increased from zero to its effective value for the first time is 9 ms, then the time required to increase from zero to half its maximum value for the first time is ..... .

(a) 3 ms (b) 6 ms (c) 12 ms (d) 18 ms

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6 An electric power has been transferred twice through a transmission line whose resistance is R, once when the station potential difference was (V) so the lost power in the line was $(P_w)_1$ and another time using an ideal step-up transformer to raise the voltage to 10 V, so the lost power in the line was $(P_w)_2$, so the ratio $\left(\frac{(P_w)_1}{(P_w)_2}\right)$ is ..... .

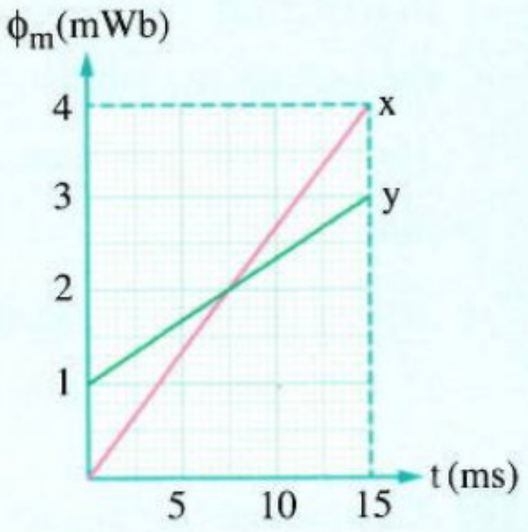
(a) $\frac{50}{1}$ (b) $\frac{100}{1}$ (c) $\frac{1}{50}$ (d) $\frac{1}{100}$

7 During the rotation of the coil of a simple motor, the magnetic dipole moment of the coil vanishes when the coil ..... .

(a) makes angle of 30° with the direction of the magnetic field
(b) makes angle of 60° with the direction of the magnetic field
(c) is parallel to the direction of the magnetic field
(d) is perpendicular to the direction of the magnetic field

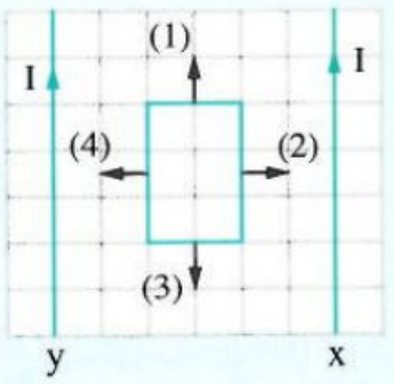
8 The opposite graph represents the relation between the magnetic flux (ϕ_m) through each of two metal rings (x, y) and time (t), if the intensities of the two induced currents in both rings are equal, the ratio between the resistances of both rings $\left(\frac{R_x}{R_y}\right)$ is ..... .

(a) $\frac{1}{5}$ (b) $\frac{5}{1}$
(c) $\frac{1}{2}$ (d) $\frac{2}{1}$



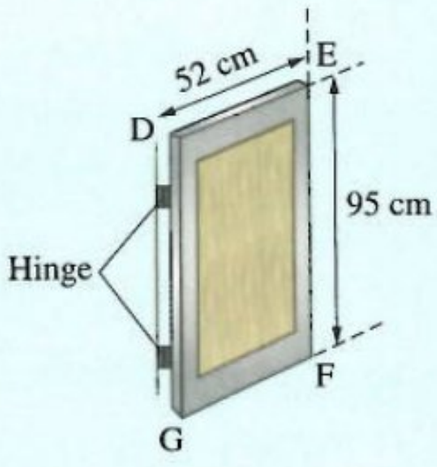
9 A rectangular metal frame is placed between two parallel straight wires (x), (y), each of them carries an electric current of intensity I in the same direction as in the opposite figure, in which of the shown directions should the frame be moved, so an induced current gets generated in the counter-clockwise direction?

(a) Direction (1) (b) Direction (2) (c) Direction (3) (d) Direction (4)



10 The opposite figure illustrates the dimensions of an aluminum frame of a vertical window DEFG whose plane is perpendicular to the horizontal component of the Earth magnetic field that has a magnitude of 1.8×10^{-5} T. If the window is opened and its plane is rotated about the vertical side DG by an angle of 90° within a time interval of 0.6 s, so the average induced emf in the frame ..... .

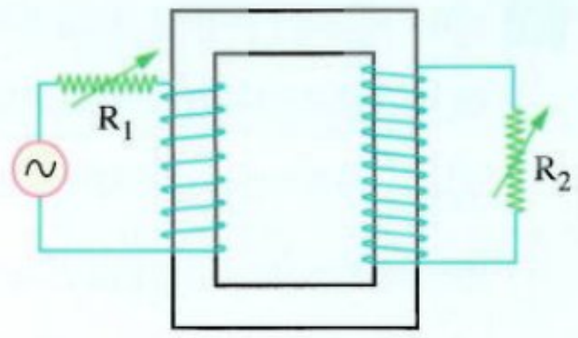
(a) 0.75×10^{-5} V (b) 1.5×10^{-5} V (c) 2.5×10^{-5} V (d) 3.2×10^{-5} V



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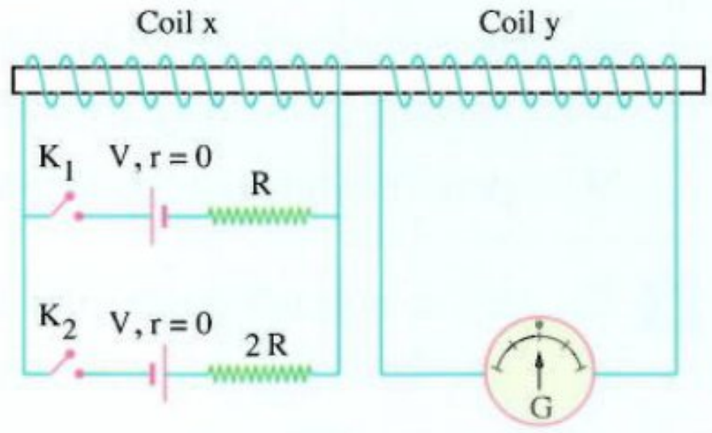
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11 In the opposite electric transformer, what happens to the value of the current in the secondary coil when :



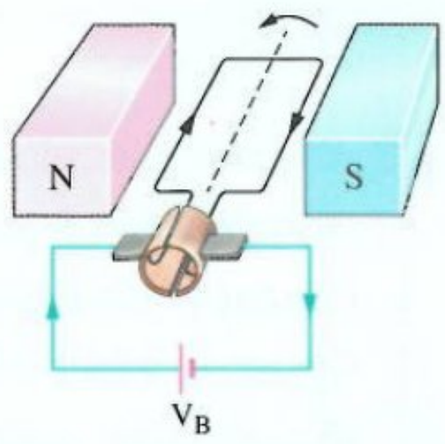
	increasing R_1 only?	increasing R_2 only?
a	Increases	Increases
b	Increases	Decreases
c	Decreases	Increases
d	Decreases	Decreases

12 The opposite figure shows two similar coils x and y, each of resistance R. At the moment of closing switch K_1, the pointer of the galvanometer which is connected to coil y gets deflected by an angle θ in a certain direction, so at the moment of closing the switch K_2 while K_1 is opened, the deflecting angle of the galvanometer pointer and its deflecting direction are ..... , ...



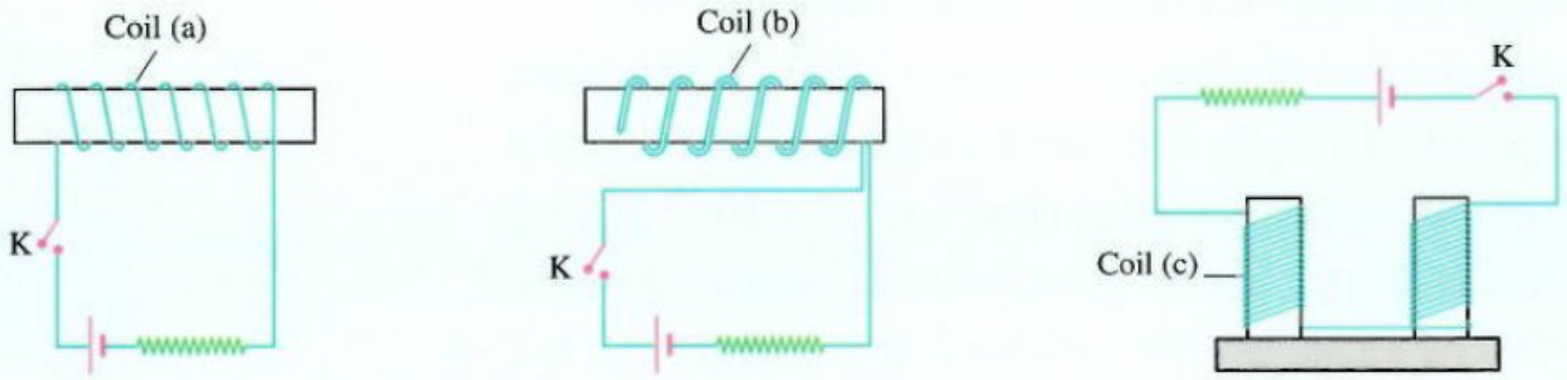
(a) greater than θ, in the same direction
(b) greater than θ, in the opposite direction
(c) less than θ, in the same direction
(d) less than θ, in the opposite direction

13 An electric motor coil of resistance (R) its terminals are connected to the terminals of an electric cell of emf (V_B) and negligible internal resistance as in the opposite figure. During the rotation of its coil between the poles of the magnet, the current intensity passing in the coil becomes ..... .



(a) $\frac{V_B}{R}$
(b) greater than $\frac{V_B}{R}$
(c) less than $\frac{V_B}{R}$
(d) the answer is indeterminable

14 Three coils a, b and c, each is connected individually in a circuit as in the following figures:

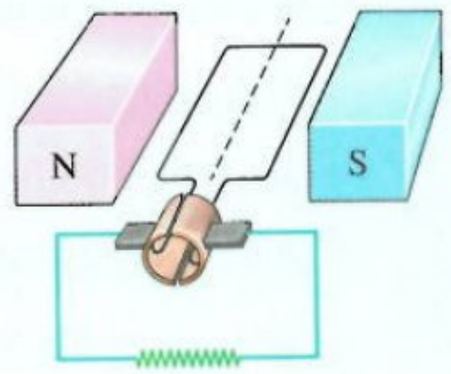


Which of these coils has no self-inductance when closing the switch K?

(a) Coil (a)
(b) Coil (b)
(c) Coil (c)
(d) None of them.

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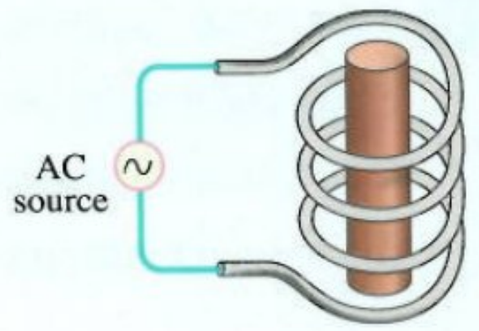
15 If the average induced emf in a dynamo coil through quarter a cycle from the shown position in the figure, equals 16 V, so its average through a complete cycle in the dynamo coil is ..... .



(a) 0 (b) 16 V

(c) 32 V (d) 64 V

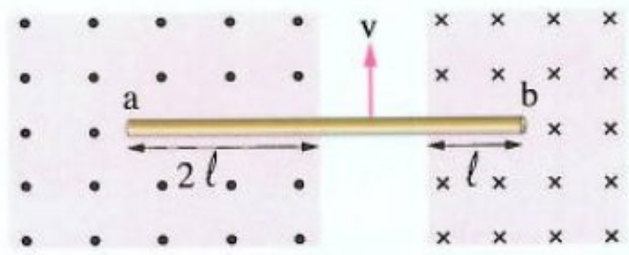
16 The opposite figure shows a coil that is coiled around a cylinder and connected to an AC source, so the eddy currents produced in the cylinder will have maximum value when the cylinder is ..... .



(a) an iron hollow cylinder (b) a plastic hollow cylinder

(c) an iron solid cylinder (d) plastic solid cylinder

17 A cylindrical metal rod ab is moving with a uniform velocity (v) in two regions of a magnetic field of density B as in the opposite figure, which of the rod terminals (a or b) has higher potential?



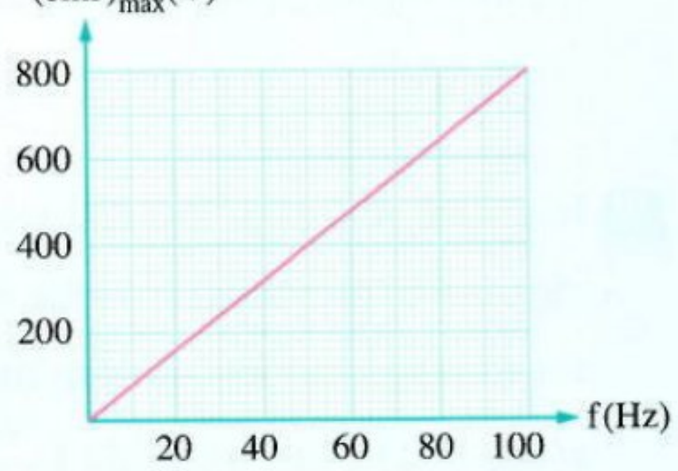
(a) Terminal a

(b) Terminal b

(c) The two terminals have the same potential

(d) The answer is indeterminable

18 A simple electric dynamo of changeable speed of rotation consists of a coil of N number of turns each of face area $\frac{4}{\pi}$ m^2. The coil rotates in a uniform magnetic field of flux density 10^{-3} T and the opposite graph represents the relation between the maximum value of the induced electromotive force $(emf)_{max}$ in the coil and the frequency (f) of the coil rotation, so the number of turns (N) of the coil equals ..... .



(a) 500 (b) 1000 (c) 1500 (d) 2000

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19 Induction coil is connected to a DC source and a switch, so the time rate of the current growth since the moment of closing the switch ..... .

(a) increases till it reaches a constant value

(b) decreases till it vanishes

(c) decreases but does not vanish

(d) does not change

---

20 Two metal frames x, y are moving with the same uniform velocity to pass through a rectangular region of a uniform magnetic field that is perpendicular to their planes until they get out completely of it. The following graphs (1), (2) represent the relation between the induced emf in each of them respectively and the time (t),

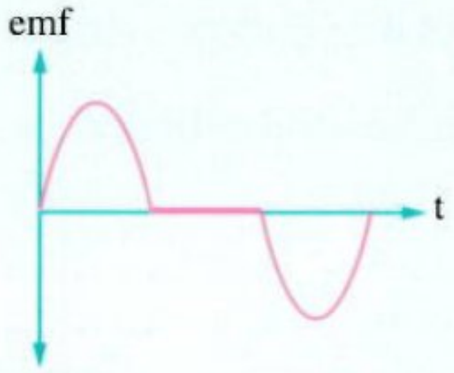


Figure (1)

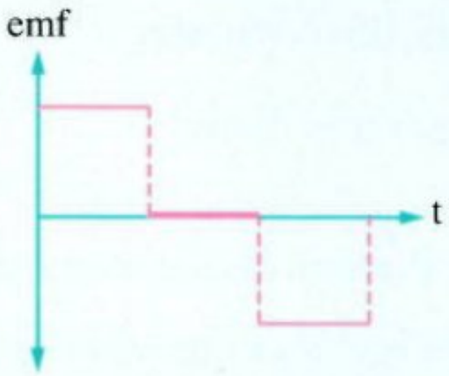


Figure (2)

so the frames (x), (y) are respectively in the form of ..... .

(a) rectangle, square

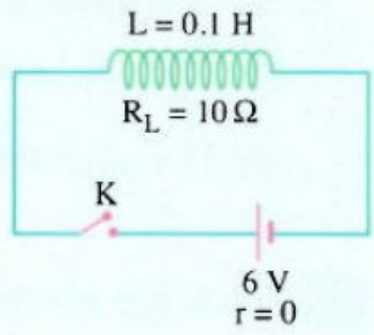
(b) square, rectangle

(c) rectangle, circle

(d) circle, rectangle

---

21 A coil of resistance 10 Ω and self-inductance 0.1 H is connected to a battery of emf 6 V and negligible internal resistance and a switch (K) as in the opposite figure, so the induced emf in the coil at the moment of closing the switch and at the moment at which the rate of the current growth is 30 A/s respectively are ..... , ...



(a) 0, 6 V (b) 6 V, 6 V (c) 0, 3 V (d) 6 V, 3 V

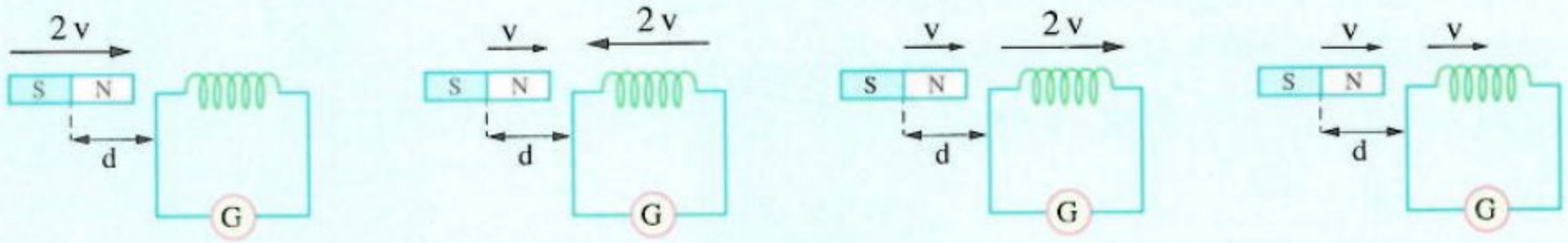
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22 In an AC dynamo, the coil takes time t to rotate $\frac{1}{3}$ cycle from the position in which its plane is perpendicular to the direction of the magnetic field, so the ratio between the average induced emf in the coil through the time interval t and the induced emf at the end of the interval t $\left(\frac{(emf)_{av}}{(emf)_{inst}}\right)$ equals ..... .

(a) $\frac{\sqrt{3}}{\pi}$ (b) $\frac{3\sqrt{3}}{2\pi}$ (c) $\frac{3\sqrt{3}}{\pi}$ (d) $\frac{\sqrt{3}}{2\pi}$

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23 The following figures show trials of a student to study the effect of the direction and the relative velocity between a magnet and a coil on the induced emf in the coil.



(1) (2) (3) (4)

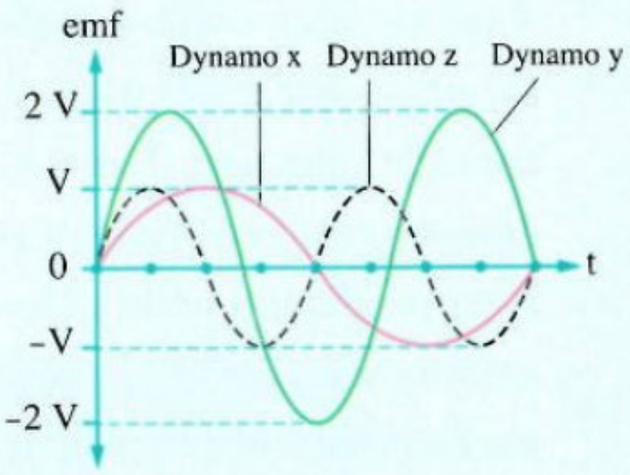
In which two of the previous figures, the deflection of the galvanometer pointer is in the same direction?

(a) Figures (1), (2) (b) Figures (1), (3) (c) Figures (2), (4) (d) Figures (1), (4)

24 An ideal transformer that its primary coil consists of 100 turns and is connected to an AC source of potential 120 V while its secondary coil that of length ℓ consists of 200 turns and coiled uniformly, what is the distance between two points on the secondary coil the potential difference between them equals 12 V ?

(a) $\frac{\ell}{5}$ (b) $\frac{\ell}{10}$ (c) $\frac{\ell}{20}$ (d) $\frac{\ell}{100}$

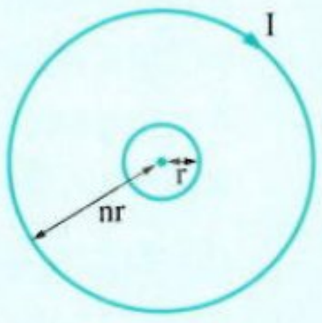
25 The opposite graph represents the generated electromotive forces (emf) from three dynamos (x, y and z) during the same time interval. If the three coils have the same cross-sectional area and are exposed to the same uniform magnetic field, the order of the coils according to their numbers of turns is .....



(a) $N_z > N_y > N_x$ (b) $N_x > N_y > N_z$

(c) $N_y > N_x = N_z$ (d) $N_y > N_x > N_z$

26 In the opposite figure, two concentric circular metal rings are in the same plane. The small ring has radius (r) and the big ring has radius (nr) and it carries a current of intensity (I). When that current vanishes in time interval (t), emf is generated in the small ring. So the mutual induction between the rings (M) is equal to ..... .



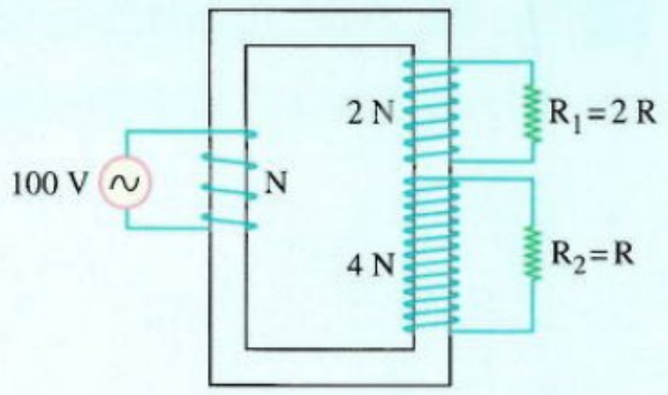
(Where: μ is the magnetic permeability of the medium of the two rings)

(a) $\frac{\mu\pi}{2} \times nr$ (b) $\frac{\mu}{2\pi} \times nr$ (c) $\frac{\mu\pi}{2} \times \frac{r}{n}$ (d) $\frac{\mu}{2\pi} \times \frac{r}{n}$

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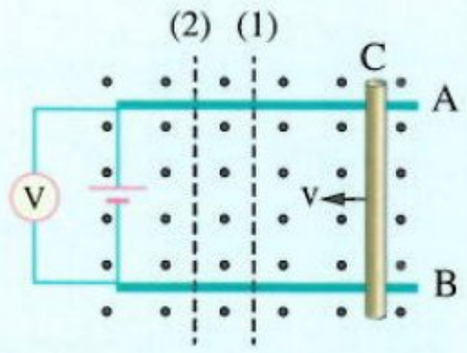
27 In the opposite figure, an ideal transformer has two secondary coils, so the ratio between the consumed powers $((P_W)_P : (P_W)_1 : (P_W)_2)$ in each of the primary coil and the resistors R_1, R_2 equal .....



(a) 9 : 1 : 1 (b) 8 : 9 : 1

(c) 9 : 1 : 8 (d) 9 : 8 : 1

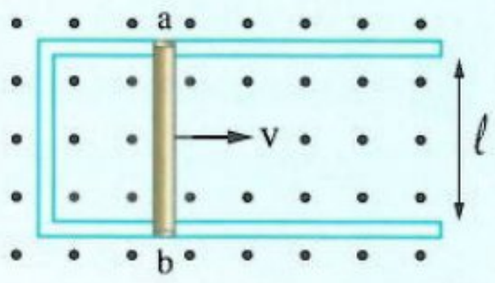
28 The opposite figure shows a metal rod C that slides on two metal rails A and B perpendicular to a uniform magnetic field with a constant velocity v. The resistance of each of the two rails (A and B) is R and they are connected to a battery of emf 10 V and internal resistance r, so if the induced emf in rod C is 4 V, the reading of the voltmeter during the motion of the rod from position (1) to position (2) ..... .



(a) gets increased (b) gets decreased (c) remains constant (d) becomes zero

Second Answer the following questions

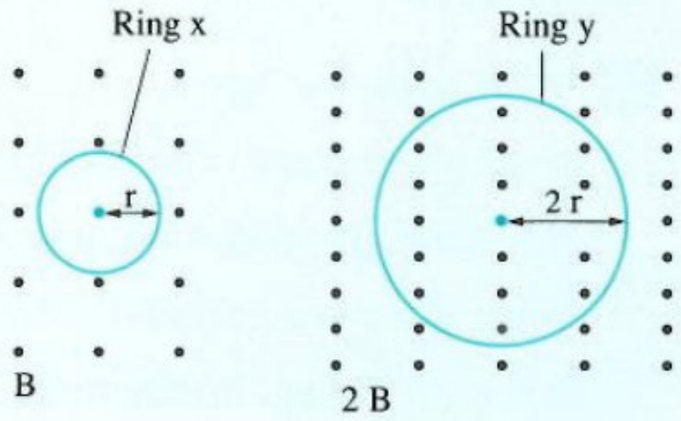
29 The opposite figure shows a rod ab of length 5 cm and resistance 4 Ω that moves with a uniform velocity 2 m/s in the plane of the page on a metal frame of negligible resistance towards the right direction. The rod and the frame are affected by a uniform magnetic field of flux density 0.4 T whose direction is perpendicular to the plane of the page, so to keep the rod ab moving with the same uniform velocity, **calculate** the value of the external force by which the rod ab should be dragged.



.....

.....

30 Two metal rings x, y are made of a wire that the resistance of a unit length of it is 1 Ω. The rings are placed perpendicular to two uniform magnetic fields of flux densities B, 2 B respectively as shown in the opposite figure, so if both fields have vanished together, **find the ratio between :**



(a) The two induced emfs in both rings $\left(\frac{emf_x}{emf_y}\right)$.

(b) The two induced current intensities in both rings $\left(\frac{I_x}{I_y}\right)$.

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